Java vs. MongoDB

Shipping Java agents to MongoDB Atlas Agent Engine

João Galego | 🧠⟩

MongoDB × Java meetup · November 2026

$ whoami

From atoms to agents

Education

MSc Physics



PgDip Forensics



PhD Cognitive Science (ABD)





Work

Lead ML Engineer



Solutions Architect



Head of AI



Critical software

We build systems that can't fail

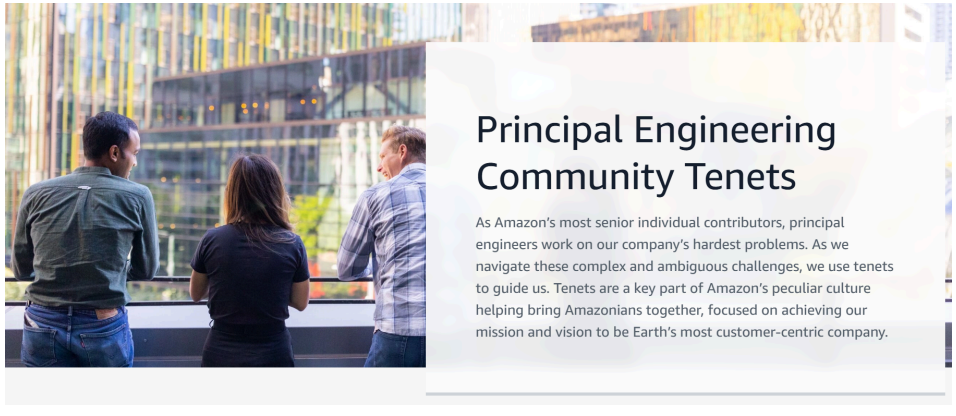
🚀 Space · ✈️ Aviation · 🚆 Railway · 🏥 Medical devices · 🚗 Automotive · 🛡️ Defence · ⚡ Energy

Since 1998 · NASA · ESA · Airbus · BMW · Alstom

Made in Coimbra

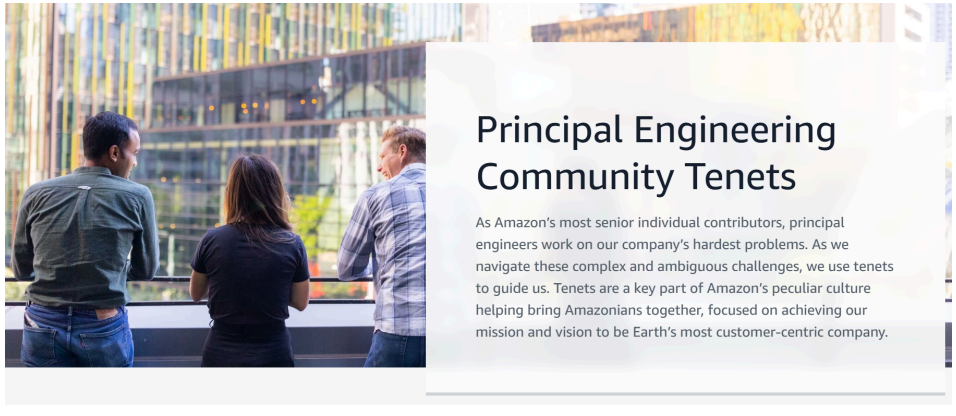


Tenets to live by



1. Exemplary practitioner
2. Technically fearless
3. Lead with empathy
4. Balanced and pragmatic
5. Illuminate and clarify
6. Flexible in approach
7. Respect what came before
8. Learn, educate, and advocate
9. Have resounding impact

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6. Flexible in approach
7. Respect what came before
8. Learn, educate, and advocate
9. Have resounding impact
10. **Have fun doing it**

Want to know what I'm up to?

```
$ finger jgalego@plan.cat
Head of AI @ Critical Software.
Invited Professor @ ISEG/ULisboa.

Now:
  * bringing ML to places where "it mostly
    works" isn't enough
  * teaching (but mostly, learning)

Todo:
  - update this .plan more often
    than once a decade

Elsewhere:
  web       https://jgalego.github.io
  code      https://github.com/JGalego
  linkedin  .../in/jgalego  (fastest HELO)
```

finger me or just scan.

Once upon a time...

Dedicated to João Mourão (MongoDB), who asked for a Disney movie

Once upon a time...



The kingdom

A Java agent dreams of Atlas

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The curse

A dragon named `java-none`

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The quest

Smuggle Java past the dragon

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Once upon a time...



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The quest

Smuggle Java past the dragon



Happily ever after

Java answers from Atlas

Dedicated to João Mourão (MongoDB), who asked for a Disney movie

PROLOGUE

Once upon a time

near miss

noun · an incident that nearly became an accident

Where the reports come from



FDA MAUDE
medical device reports
`openFDA API`

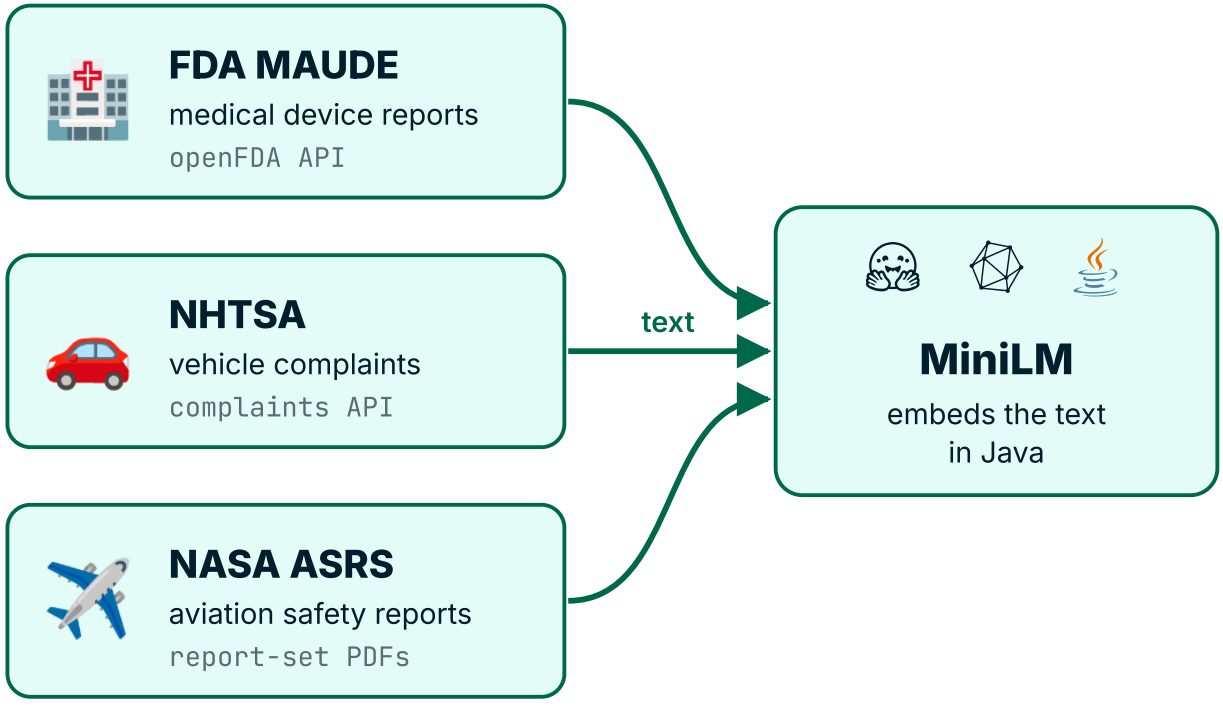


NHTSA
vehicle complaints
`complaints API`

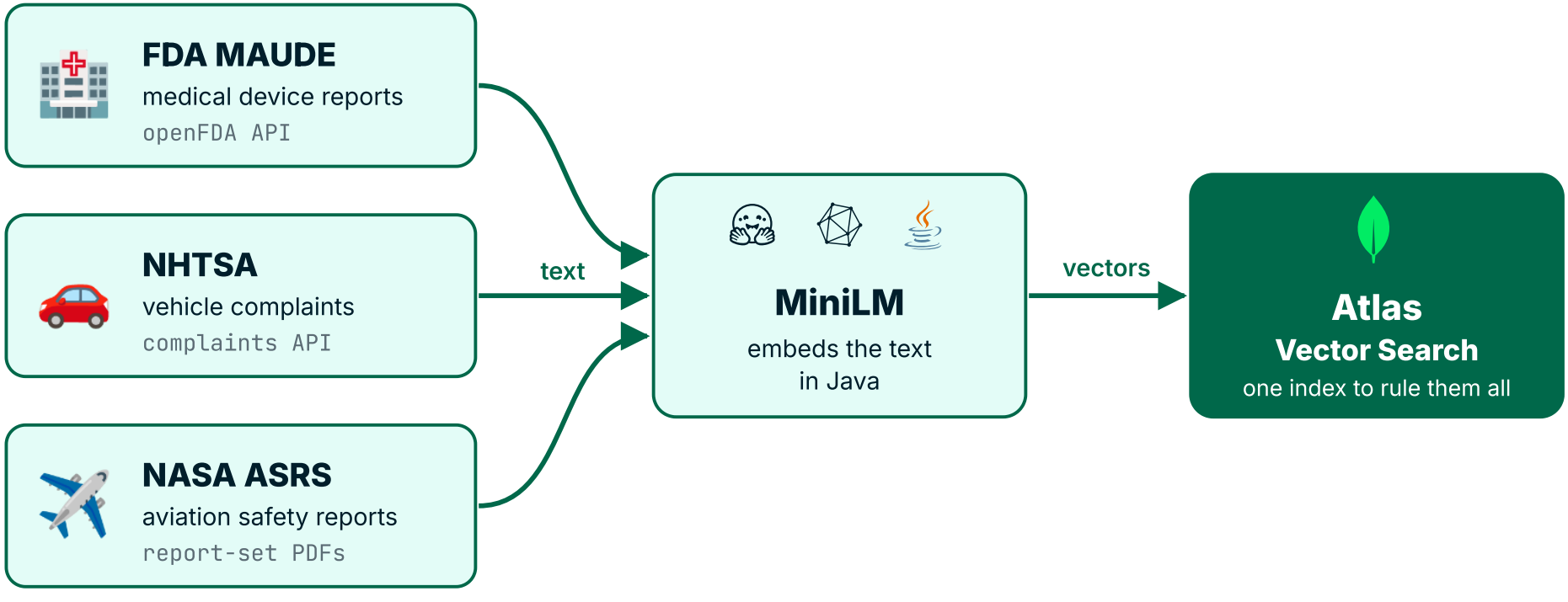


NASA ASRS
aviation safety reports
`report-set PDFs`

Where the reports come from



Where the reports come from



Look ma, no console

AtlasVectorStore.java

```
var definition = new Document("fields", List.of(
    new Document("type", "vector").append("path", "embedding")
        .append("numDimensions", dims)  // 384 MiniLM, 1024 Voyage
        .append("similarity", "cosine"),
    new Document("type", "filter").append("path", "kind"),
    new Document("type", "filter").append("path", "dataset_id")));

collection.createSearchIndexes(List.of(new SearchIndexModel(
    "near_miss", definition, SearchIndexType.vectorSearch())));
```

Look ma, no console

AtlasVectorStore.java

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collection.createSearchIndexes(List.of(new SearchIndexModel(
    "near_miss", definition, SearchIndexType.vectorSearch())));
```

```
vectorSearch(fieldPath("embedding"), query, "near_miss", limit,
    approximateVectorSearchOptions(candidates)
        .filter(eq("dataset_id", "nasa-asrs")))
```

Look ma, no console

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```
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Clicks don't survive a `git clone`.

Search by meaning

Search by meaning

🧑 "engine lost power during climb"

Search by meaning

🧑 "engine lost power during climb"

```
0.533  M-4, Takeoff / Launch. Pilot flying Maule M4 aircraft
       reported engine loss of power inflight…

0.479  B737-700, Climb. Air carrier Captain reported an engine
       failure during climb…

0.42   Aeronca Champion, Commuter and GA Icing Incidents…
```

Search by meaning

🧑 "engine lost power during climb"

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```

Different words, similar meaning.

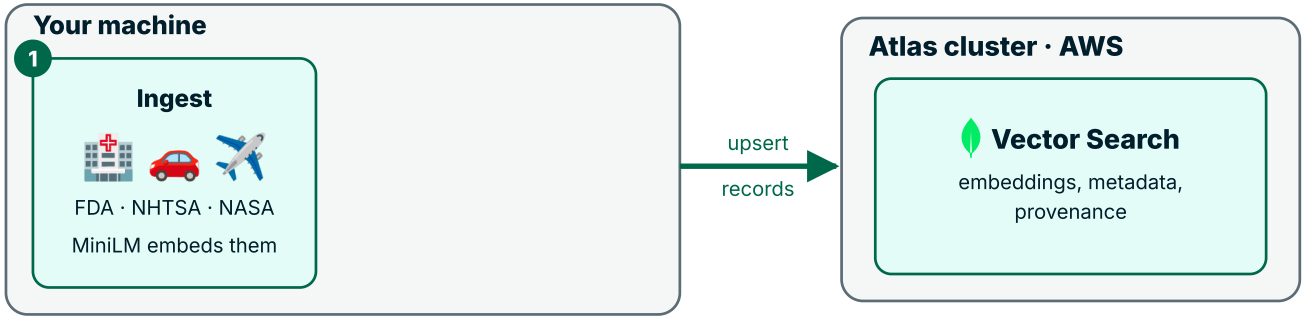
Spoiler: the happy ending

"The Road goes ever on and on / Down from the door where it began." — J.R.R. Tolkien, *The Fellowship of the Ring*

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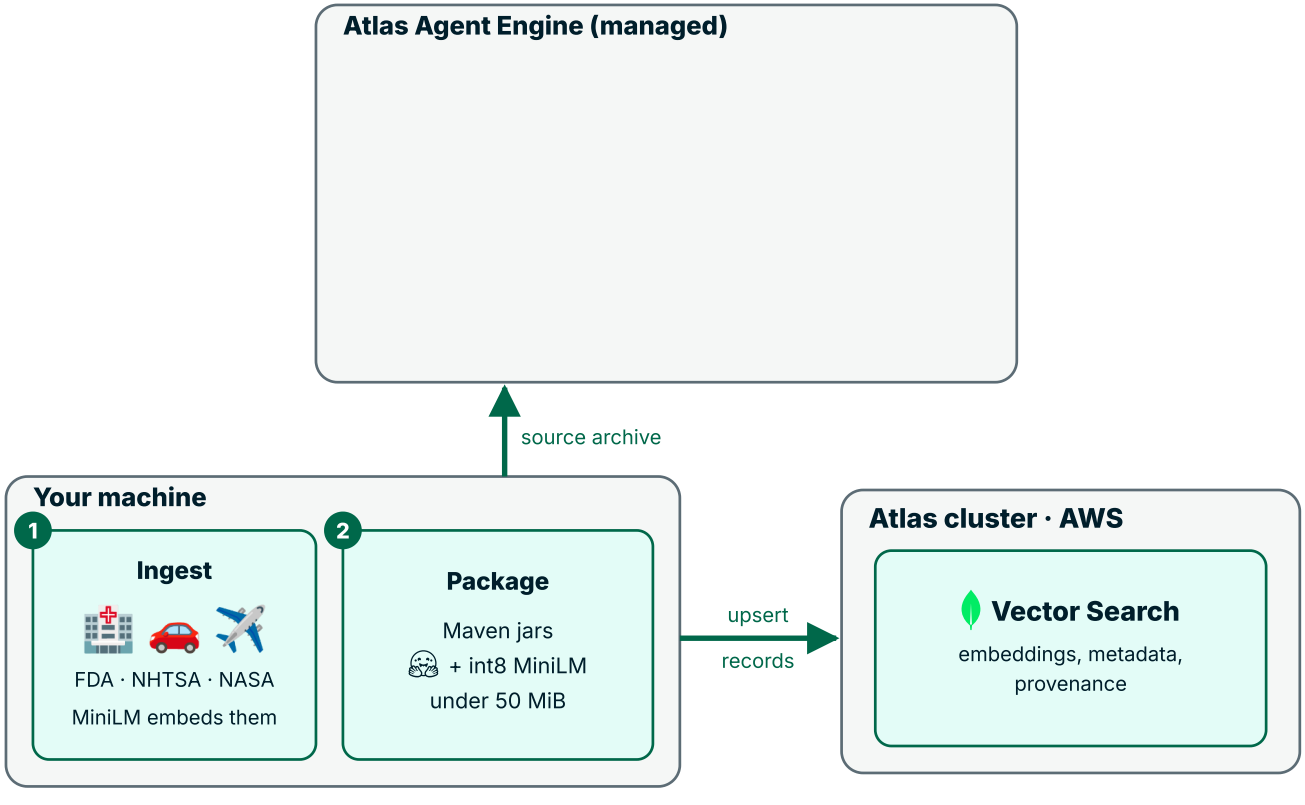
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1 **Ingest**: embed the public reports and upsert them into Atlas



Spoiler: the happy ending

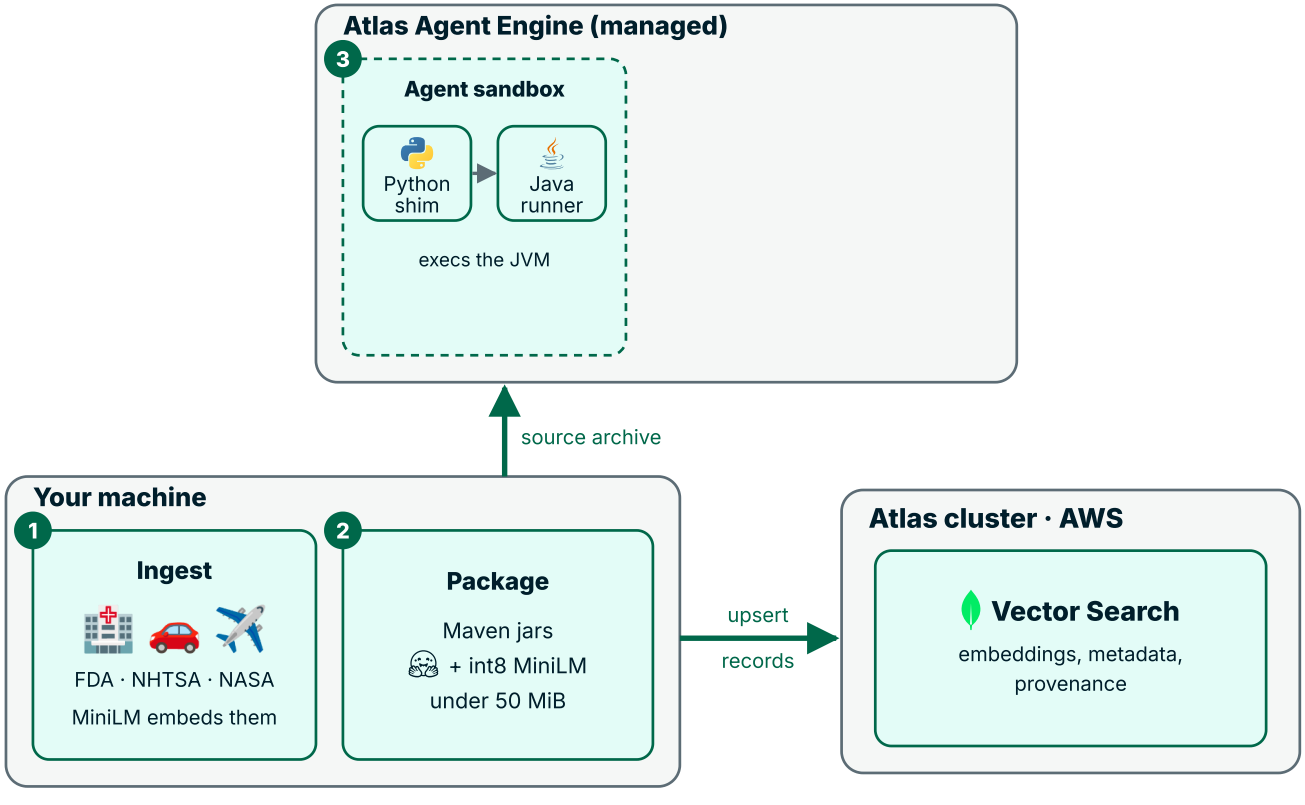
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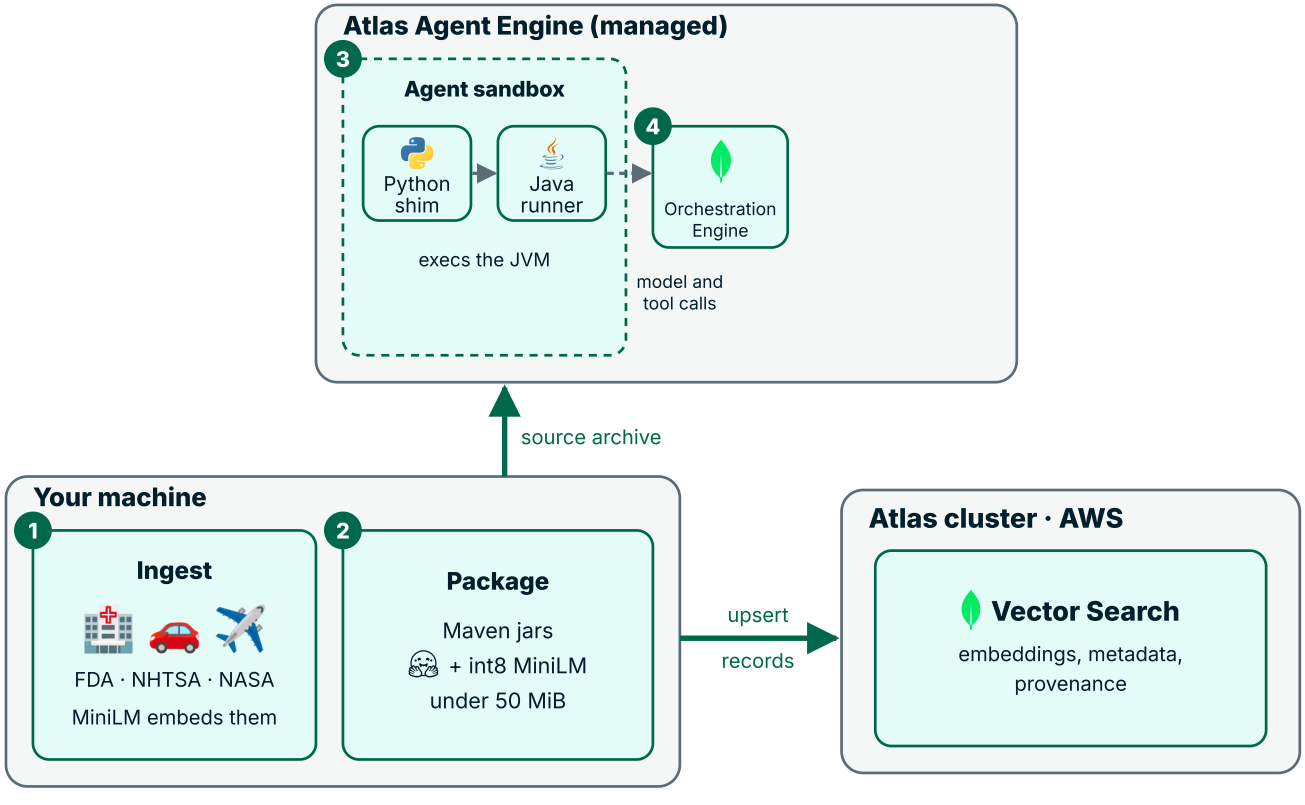
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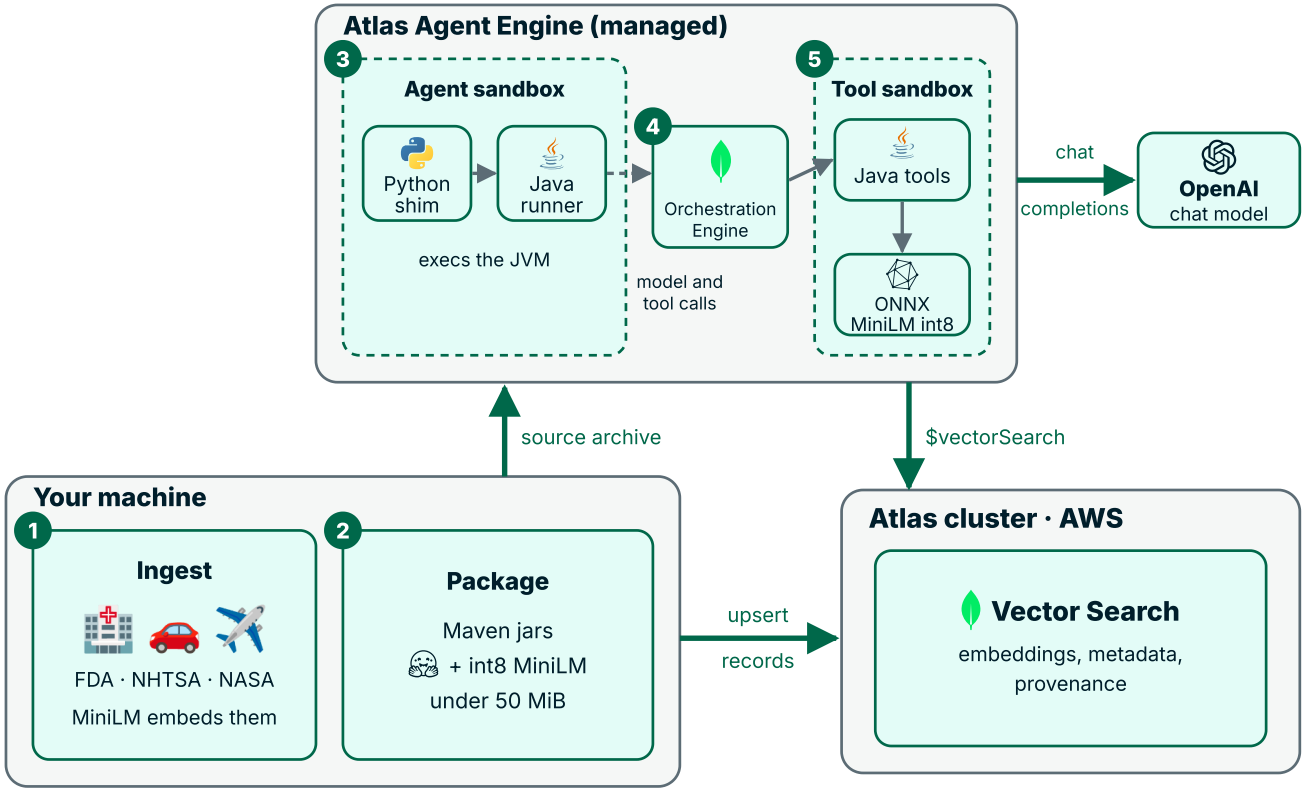
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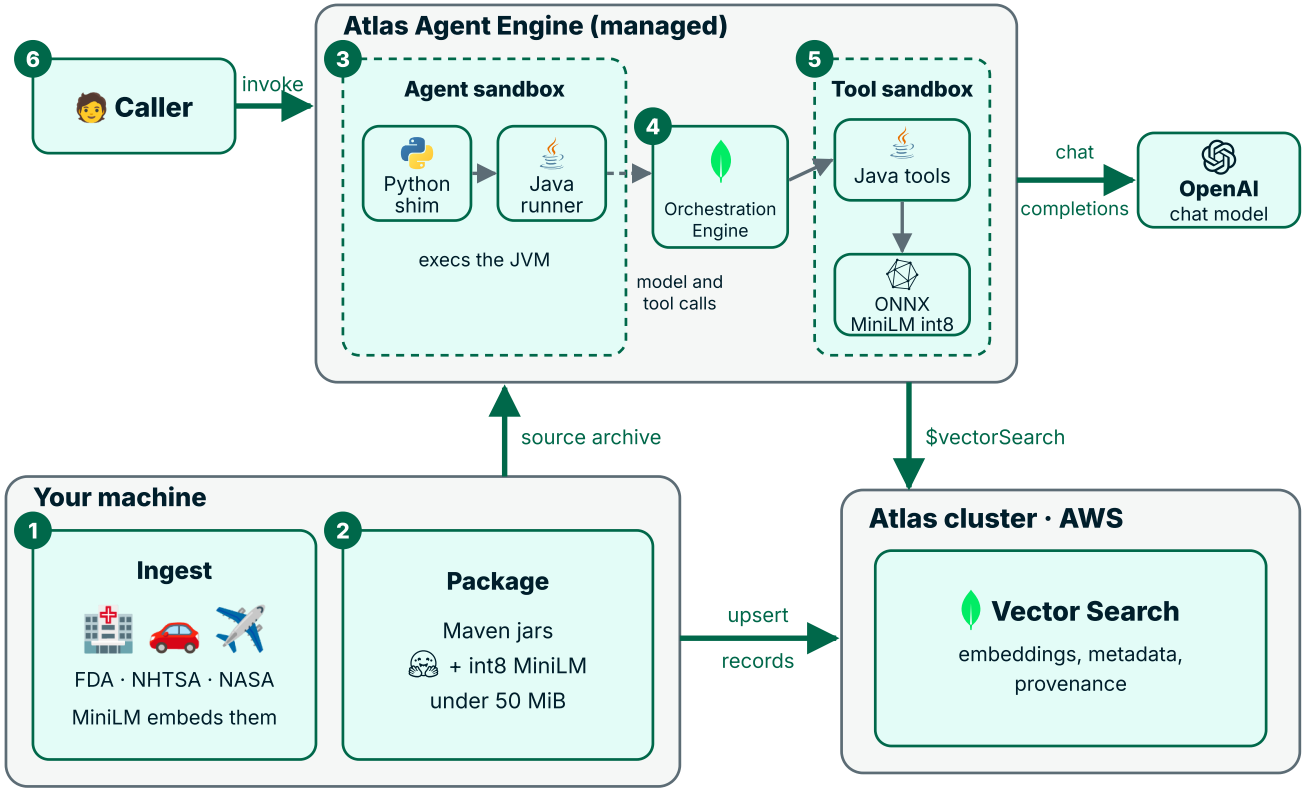
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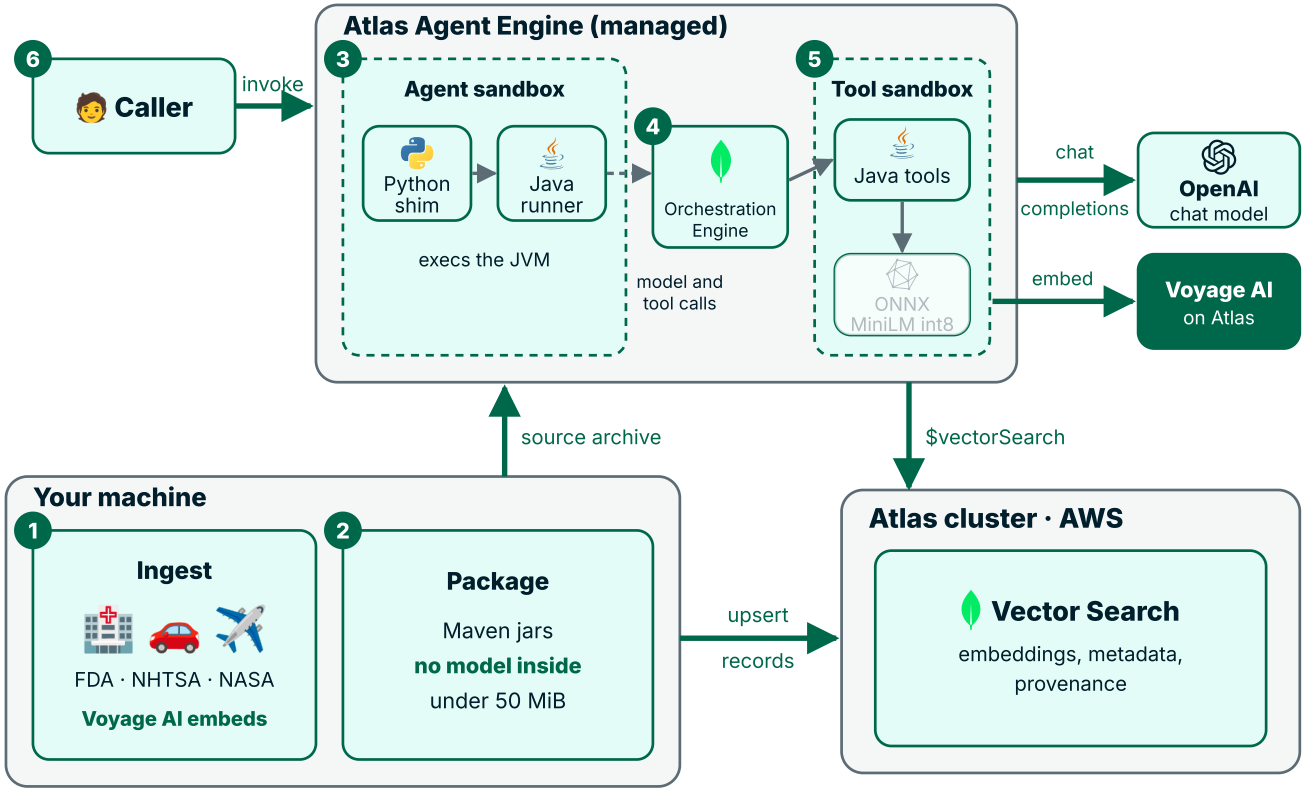
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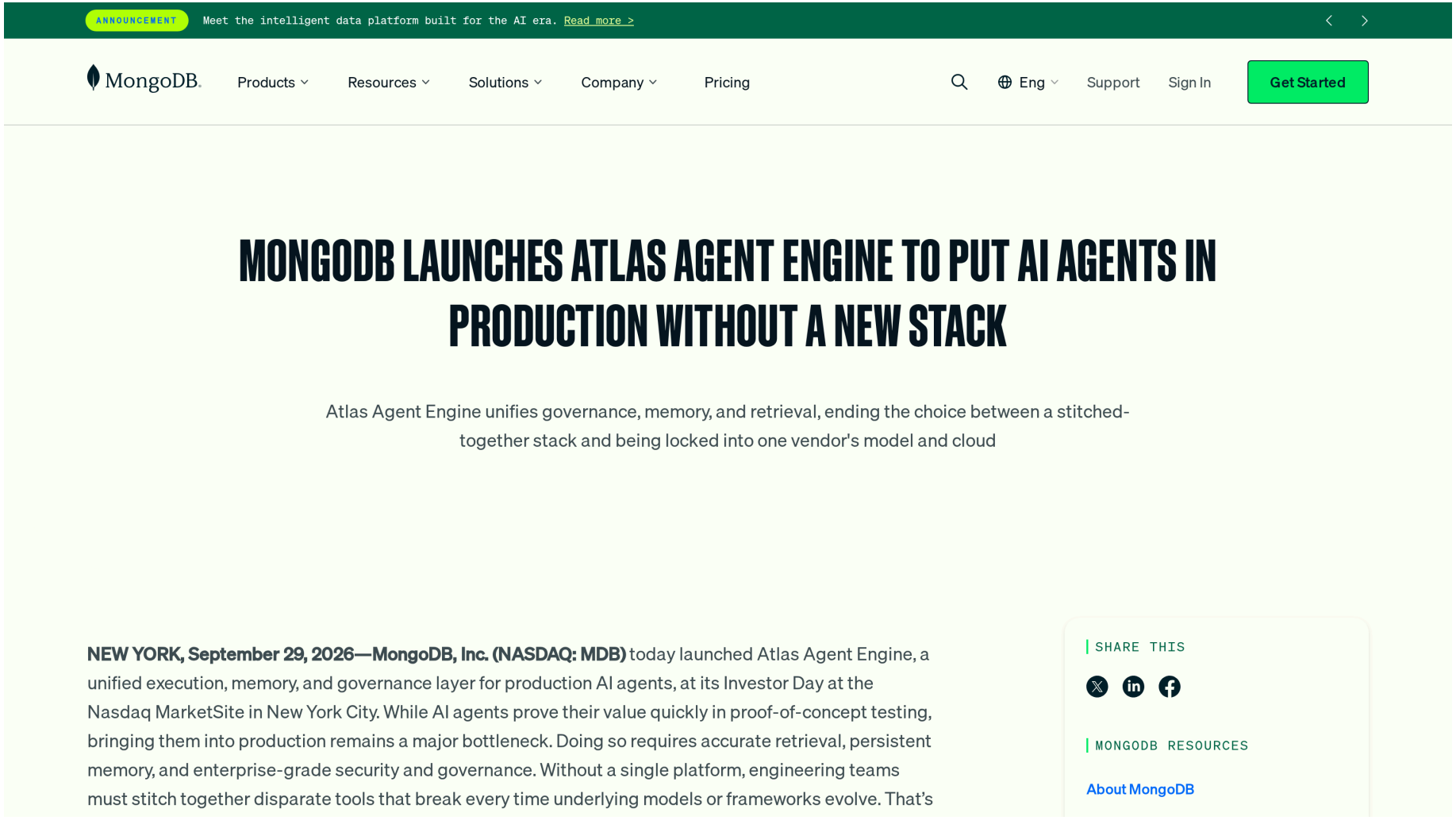
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✨ **Or flip one switch**: Voyage AI on Atlas does the embedding

CHAPTER 1

Dude, where's my Java SDK?

The press release



ANNOUNCEMENT Meet the intelligent data platform built for the AI era. Read more >

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MONGODB LAUNCHES ATLAS AGENT ENGINE TO PUT AI AGENTS IN PRODUCTION WITHOUT A NEW STACK

Atlas Agent Engine unifies governance, memory, and retrieval, ending the choice between a stitched-together stack and being locked into one vendor's model and cloud

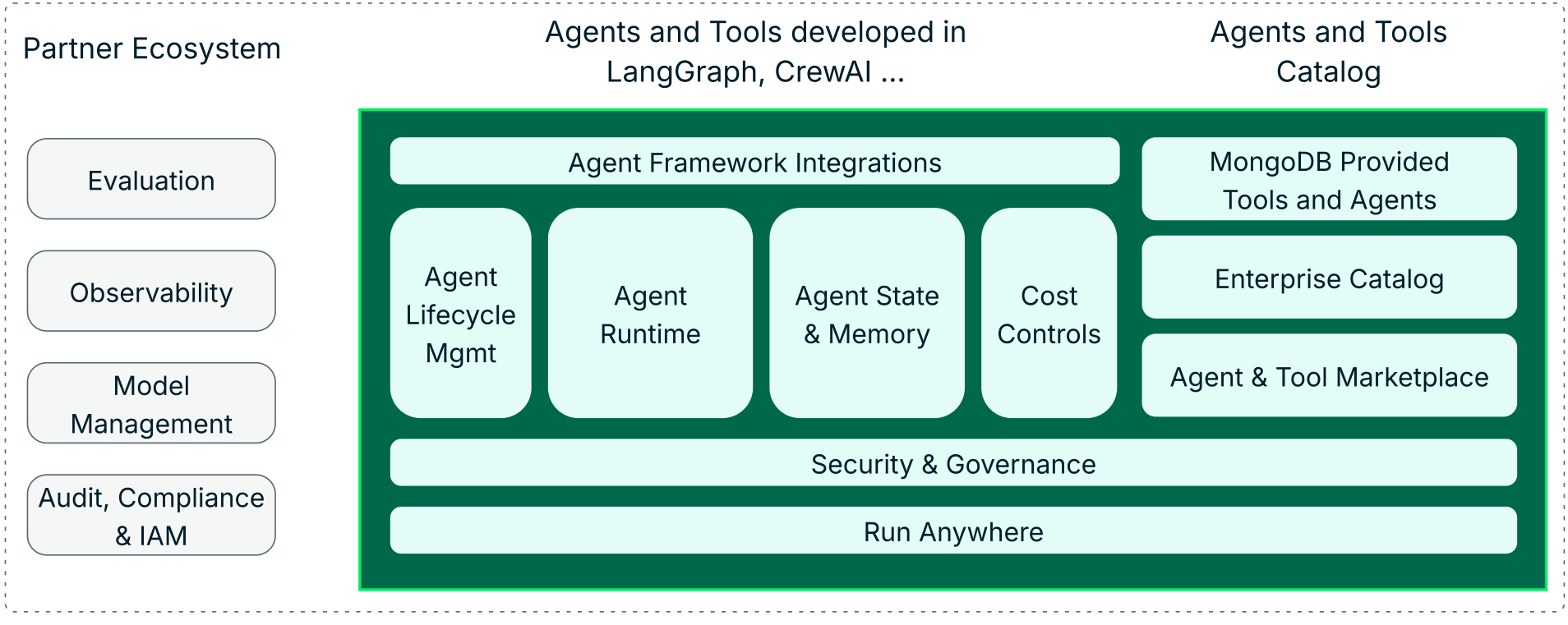
NEW YORK, September 29, 2026—MongoDB, Inc. (NASDAQ: MDB) today launched Atlas Agent Engine, a unified execution, memory, and governance layer for production AI agents, at its Investor Day at the Nasdaq MarketSite in New York City. While AI agents prove their value quickly in proof-of-concept testing, bringing them into production remains a major bottleneck. Doing so requires accurate retrieval, persistent memory, and enterprise-grade security and governance. Without a single platform, engineering teams must stitch together disparate tools that break every time underlying models or frameworks evolve. That's

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About MongoDB

Introducing the MongoDB Agentic Platform



⚠️ Atlas Agent Engine is in **public preview**: not yet recommended for production.

Introducing the MongoDB Agentic Platform



SHUT UP WITH THE SALES PITCH.

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Introducing the MongoDB Agentic Platform

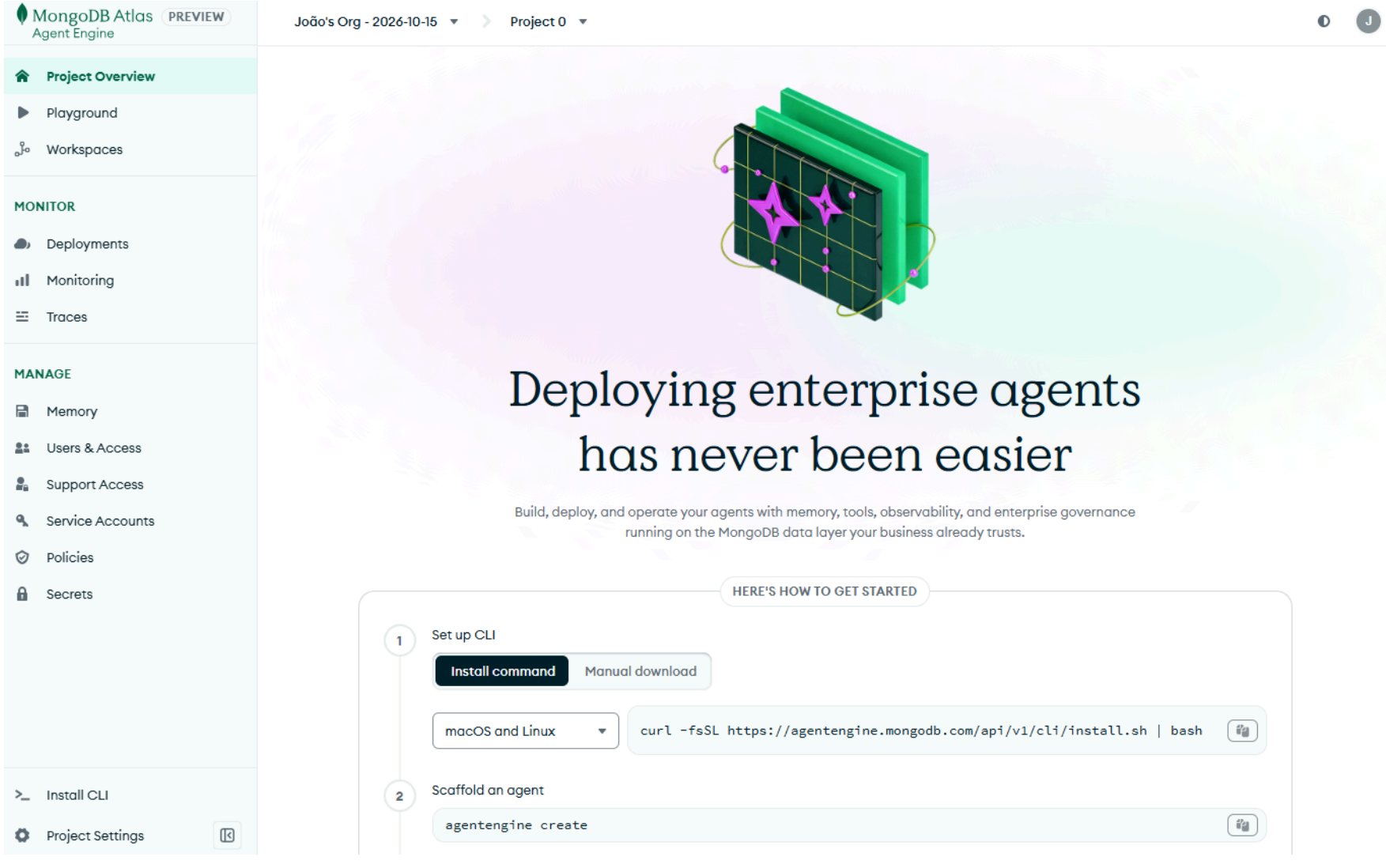


SHUT UP WITH THE SALES PITCH.

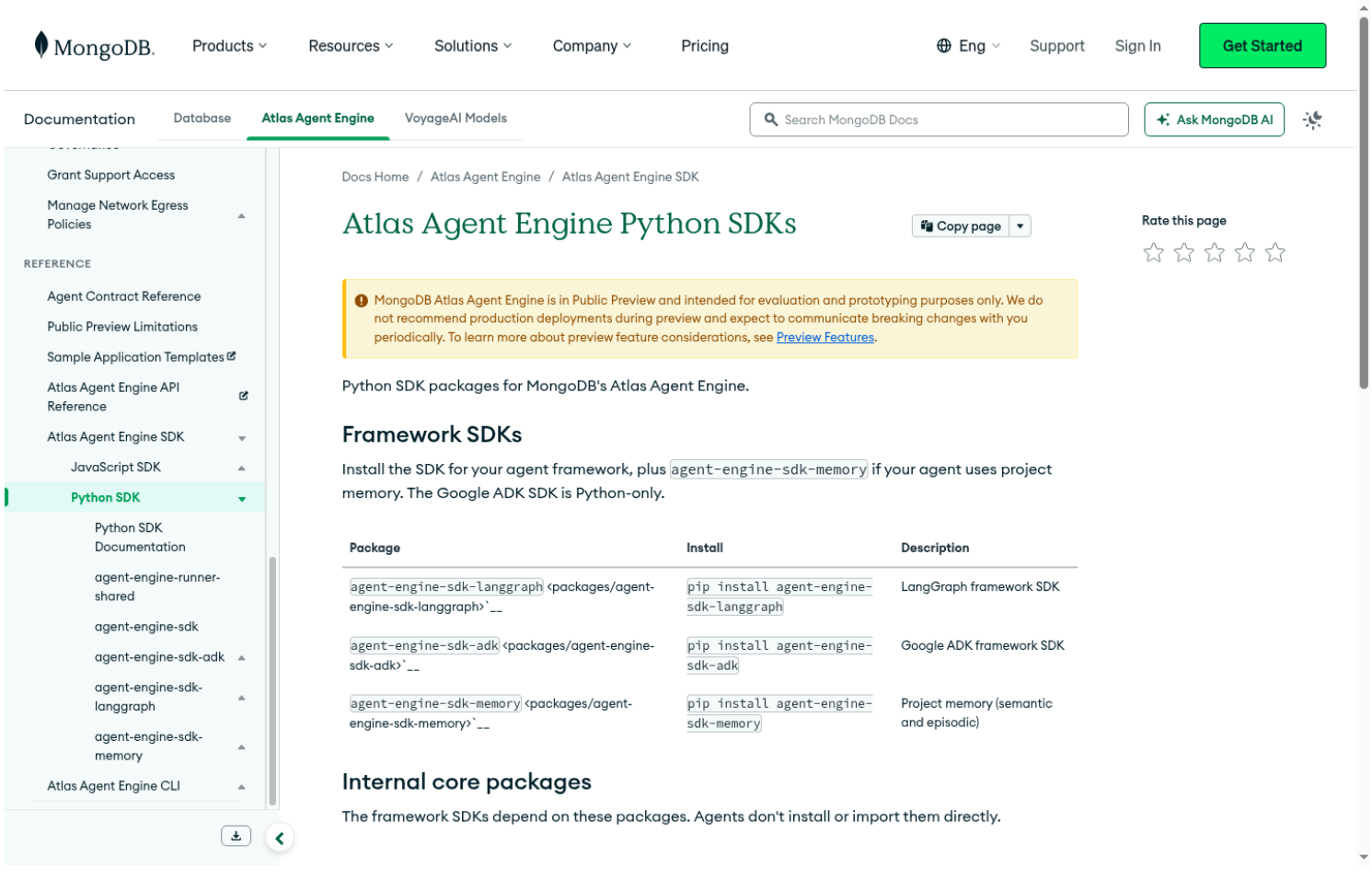
LET ME TRY IT.

⚠️ Atlas Agent Engine is in **public preview**: not yet recommended for production.

The front door



Python SDK: 200 ✅



Java SDK: 404 ❌

MongoDB. Products Resources Solutions Company Pricing Eng Support Sign In Get Started

Search MongoDB Docs Ask MongoDB AI



Sorry, we can't find that page.

The page with the URL "https://www.mongodb.com/docs/agentengine/sdk/java/" does not exist. It might have been moved or deleted.

The MongoDB Assistant suggests the following pages instead:

- Atlas Agent Engine TypeScript SDKs
  https://mongodb.com/docs/agentengine/sdk/javascript
- Table of Contents
  https://mongodb.com/docs/agentengine/sdk/python/packages/agent-engine-sdk-langgraph/docs/api
- @mongodb-js/agent-engine-sdk-langgraph
  https://mongodb.com/docs/agentengine/sdk/javascript/packages/agent-engine-sdk-

The CLI says no

```
$ agentengine init

Error: unsupported language/framework combination "java-none"

(supported: python-langgraph, python-adk,
            python-openai-agents, typescript-langgraph)
```

worst name ever!

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```

This talk is about this error message.

The repo has no Java

```
agent-engine-client-libraries
```

The repo has no Java

```
agent-engine-client-libraries
 |
 ├── python/        ✅  SDK, memory, runner, LangGraph, ADK, OpenAI
```

The repo has no Java

```
agent-engine-client-libraries
  |
  ├── python/       ✅  SDK, memory, runner, LangGraph, ADK, OpenAI
  |
  ├── javascript/   ✅  SDK, memory, runner, LangGraph
```

The repo has no Java

```
agent-engine-client-libraries
  |
  ├── python/       ✅  SDK, memory, runner, LangGraph, ADK, OpenAI
  |
  ├── javascript/   ✅  SDK, memory, runner, LangGraph
  |
  └── java/         😐
```

The plan

The plan

How hard can it be?

CHAPTER 2

But it works on my machine

core SDK → memory → LangChain4j → runner → LangGraph4j → ☁️

The core SDK

core SDK → memory → LangChain4j → runner → LangGraph4j → ☁️

The core SDK

- 📘 The models live in TypeScript (zod schemas)

core SDK → memory → LangChain4j → runner → LangGraph4j → ☁️

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core SDK → memory → LangChain4j → runner → LangGraph4j → ☁️

The core SDK

- 📘 The models live in TypeScript (zod schemas)
- ☕ Port them to Java **records**, same wire format
- 🤖 Just **Copilot** your way through
- 🟢 19 model tests, green on the first run

Zod in, record out

models.ts

```ts
export const DocumentBlockSchema = z.object({
  type: z.literal("document").default("document"),
  url: z.string(),
  mime_type: z.string().optional(),
  filename: z.string().optional(),
});
```

Zod in, record out

models.ts

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export const DocumentBlockSchema = z.object({
  type: z.literal("document").default("document"),
  url: z.string(),
  mime_type: z.string().optional(),
  filename: z.string().optional(),
});
```

DocumentBlock.java

```
public record DocumentBlock(
    String url, String mimeType, String filename)
    implements ContentBlock {
  public DocumentBlock {
    Objects.requireNonNull(url, "url");  // required
  }
}
```

core SDK → **memory** → LangChain4j → runner → LangGraph4j → ☁️

The memory SDK

core SDK → **memory** → LangChain4j → runner → LangGraph4j → ☁️

The memory SDK

- 🐍 **Memory Server**: a Python web service

core SDK → **memory** → LangChain4j → runner → LangGraph4j → ☁️

The memory SDK

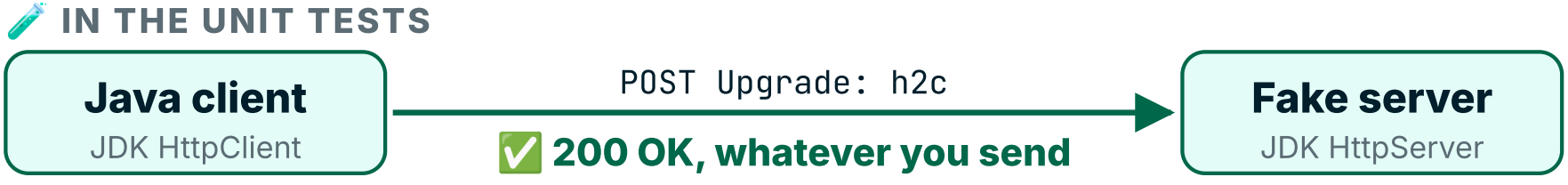
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- ☕ **My SDK**: the JDK's `HttpClient`

core SDK → **memory** → LangChain4j → runner → LangGraph4j → ☁️

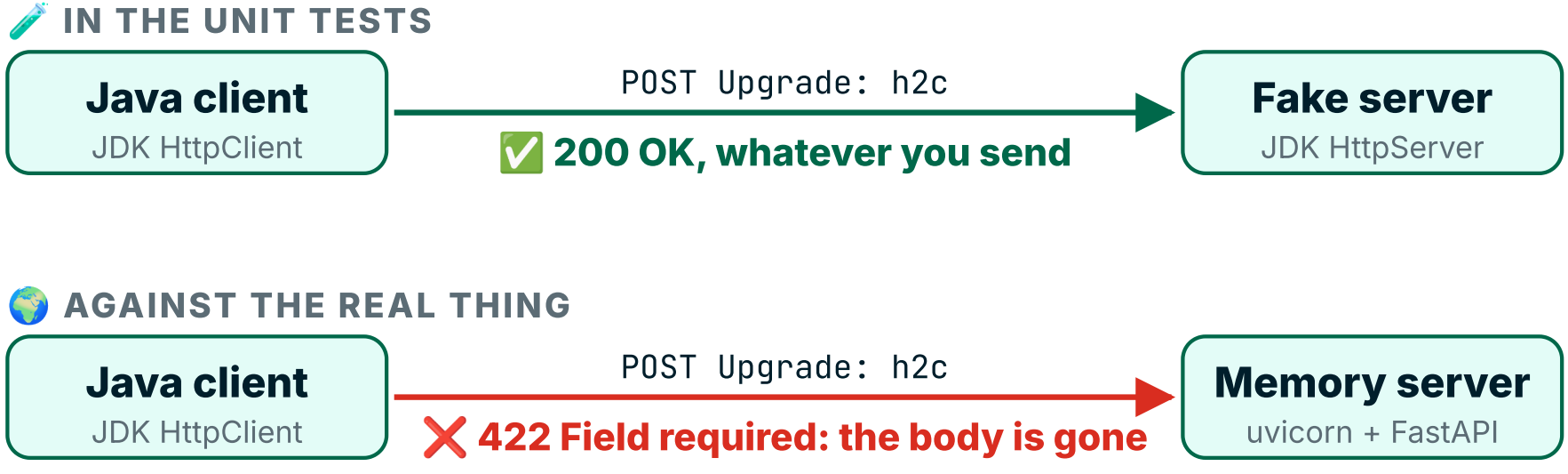
The memory SDK

- 🐍 **Memory Server**: a Python web service
- ☕ **My SDK**: the JDK's `HttpClient`
- 🧪 **The tests**: a fake server (`HttpServer`)

Problem #1: Tests pass, writes fail



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What's h2c?

HTTP/2 *over cleartext*. No TLS, just plain `http://`.

References: JDK 17 `HttpClient.Builder.version()`,
RFC 9110 §7.8 (Upgrade), §15.2.2 (101), RFC 7540 §3.2 (h2c)

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```
POST /… HTTP/1.1
Connection: Upgrade, HTTP2-Settings
Upgrade: h2c                                  ← can we talk HTTP/2 instead?
```

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😎 "Sure, switching."
`101` Switching Protocols

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POST /… HTTP/1.1
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Upgrade: h2c                                  ← can we talk HTTP/2 instead?
```

😎 "Sure, switching."
`101` Switching Protocols

🙄 "Nah, HTTP/1.1 it is."
`200` OK, as if you never asked

References: JDK 17 `HttpClient.Builder.version()`,
RFC 9110 §7.8 (Upgrade), §15.2.2 (101), RFC 7540 §3.2 (h2c)

The request body never arrived

```
→ POST  Connection: Upgrade, HTTP2-Settings
        Upgrade: h2c
        (body: the memory write)

← 422   {"detail": [{"loc": ["body"], "msg": "Field required"}]}
```

Why?

Why?

☕ **JDK client**: "Can we talk HTTP/2? Also, here's my body."

POST + `Upgrade: h2c` + the memory write

Why?

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POST + `Upgrade: h2c` + the memory write

🦄 **uvicorn**: "Upgrade? Body? What body?"

drops the body of the upgraded request

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422 it sees an empty body

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drops the body of the upgraded request

🐍 **FastAPI**: "Field required."

`422` it sees an empty body

🧪 **Fake test server**: "Upgrade? Body? Sure, whatever."

`200` it accepted the upgrade, so every test passed

The fix

Http.java

```
 HttpClient.newBuilder()
+    .version(HttpClient.Version.HTTP_1_1)  // h2c ate the body
     .connectTimeout(connectTimeout)
     .build();
```

The fix

Http.java

```
 HttpClient.newBuilder()
+    .version(HttpClient.Version.HTTP_1_1)  // h2c ate the body
     .connectTimeout(connectTimeout)
     .build();
```

MemoryClientTest.java

```
@Test
void plainHttpRequestsDoNotAskForAnH2cUpgrade() {
  client().build().getDistinctDomains(null);
  assertNull(server.only().header("Upgrade"));
}
```

The fix

Http.java

```
 HttpClient.newBuilder()
+    .version(HttpClient.Version.HTTP_1_1)  // h2c ate the body
     .connectTimeout(connectTimeout)
     .build();
```

MemoryClientTest.java

```
@Test
void plainHttpRequestsDoNotAskForAnH2cUpgrade() {
  client().build().getDistinctDomains(null);
  assertNull(server.only().header("Upgrade"));
}
```

Fake servers only know what you taught them.

Problem #2: Embedding rate limit exceeded

```
searchSemantic

Error: Embedding service rate limit exceeded
```

Whose limit is it?

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guess #1. Right, but nobody believed it

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told the human. Confidently. Wrong.

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Voyage rate limits, usage tier 1

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told the human. Confidently. Wrong.

🪵 **The log**: "You have not yet added your payment method..."

"...reduced rate limits of `3 RPM` and 10K TPM"

The fix

The fix

Read the log before you theorize.

CHAPTER 3

Tap the Wire

core SDK → memory → **LangChain4j** → runner → LangGraph4j → ☁️

The LangChain4j adapter

core SDK → memory → **LangChain4j** → runner → LangGraph4j → ☁️

The LangChain4j adapter

- 🔌 Any LangChain4j chat model becomes the platform's LLM

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- 🟢 All green. Models: done

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The LangChain4j adapter

- 🔌 Any LangChain4j chat model becomes the platform's LLM
- 🧪 45 offline tests, 6 live against OpenAI
- 🟢 All green. Models: done
- 🏃 Next up: the **runner**

Any model, one adapter

LangChain4jLiveTest.java

```
var llm = LangChain4jLLM.builder()
    .chatModel(OpenAiChatModel.builder()
        .apiKey(key).modelName("gpt-4o-mini").build())
    .build();
```

Any model, one adapter

LangChain4jLiveTest.java

```
var llm = LangChain4jLLM.builder()
    .chatModel(OpenAiChatModel.builder()
        .apiKey(key).modelName("gpt-4o-mini").build())
    .build();
```

```
LLMResponse response = llm.invoke(
    List.of(Message.user("Reply with exactly the word: pong")));
// response.content() → "pong", response.usage() → tokens
```

The docs say no Java

core SDK → memory → LangChain4j → **runner** → LangGraph4j → ☁️

> *"Agents do not implement their own HTTP endpoint."* — *the docs*

The docs say no Java

core SDK → memory → LangChain4j → **runner** → LangGraph4j → ☁️

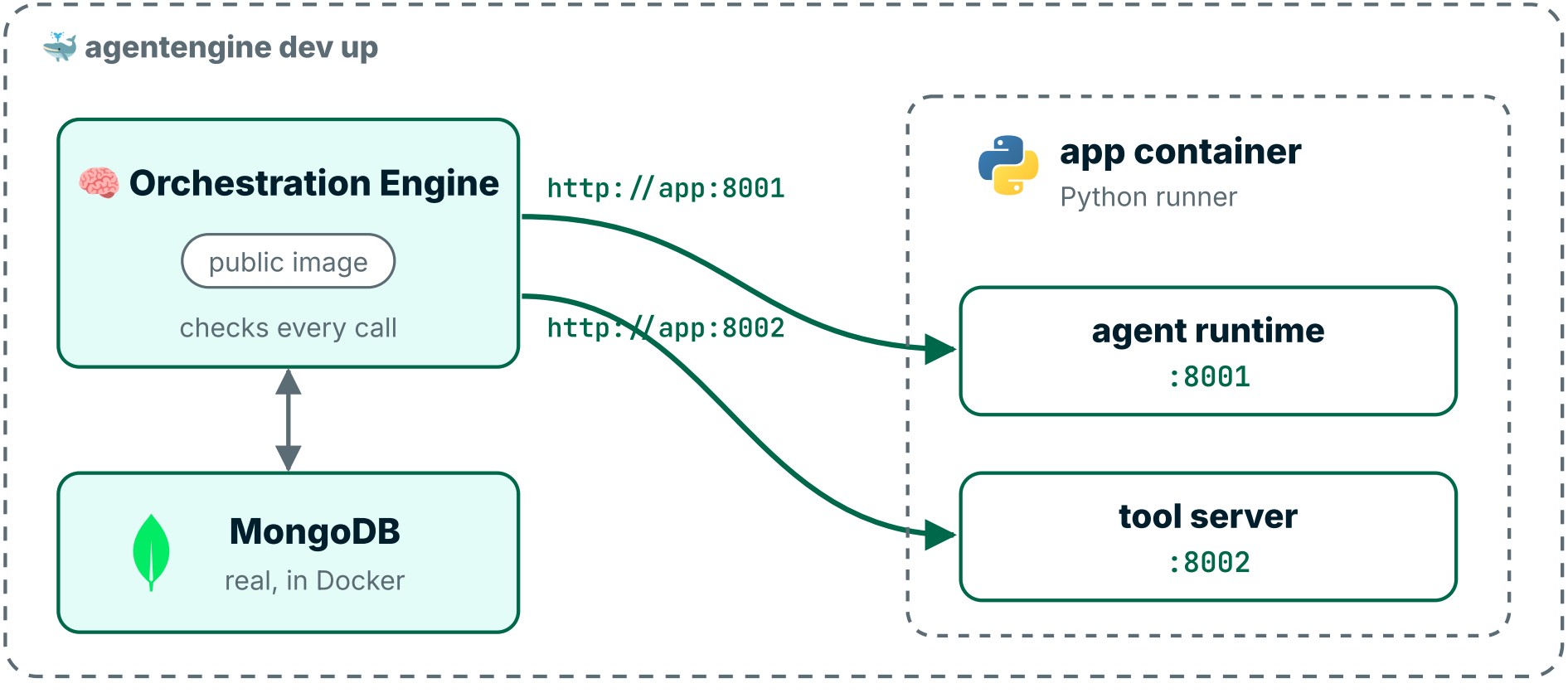
> *"Agents do not implement their own HTTP endpoint."* — *the docs*

So: no Java on the managed platform. 😬

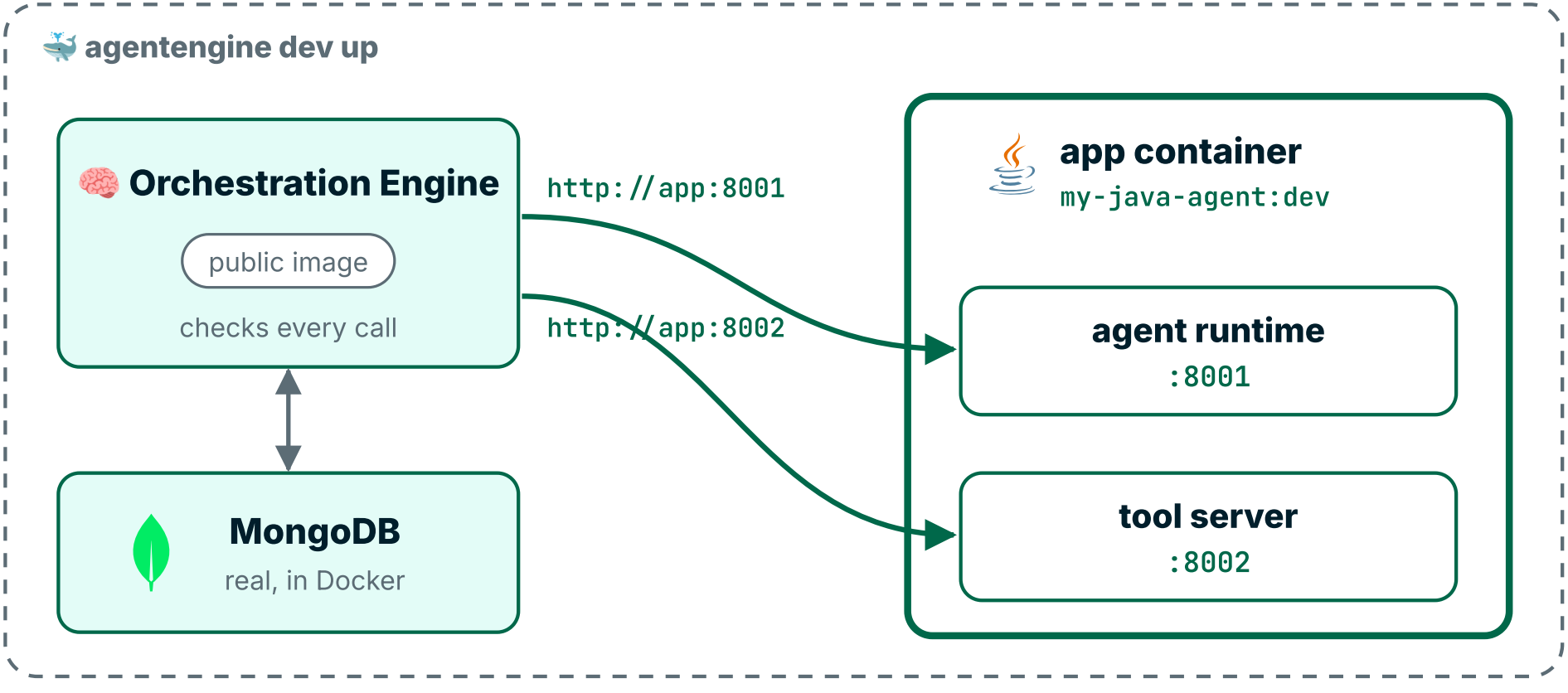
The way in

Then how do they talk? 🤔

The local stack



The local stack



Swap one container

The local stack's Orchestration Engine is a **public image**. Swap one container:

```
services:
  app:
    image: my-java-agent:dev
    build: !reset null      # bye, Python
    volumes: !reset []
```

Watching the baseline

```
$ agentengine invoke "…"

runtime_lost
```

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```
$ agentengine invoke "…"

runtime_lost
```

The app's log: *"durable Memory identity is incomplete"*

Watching the baseline

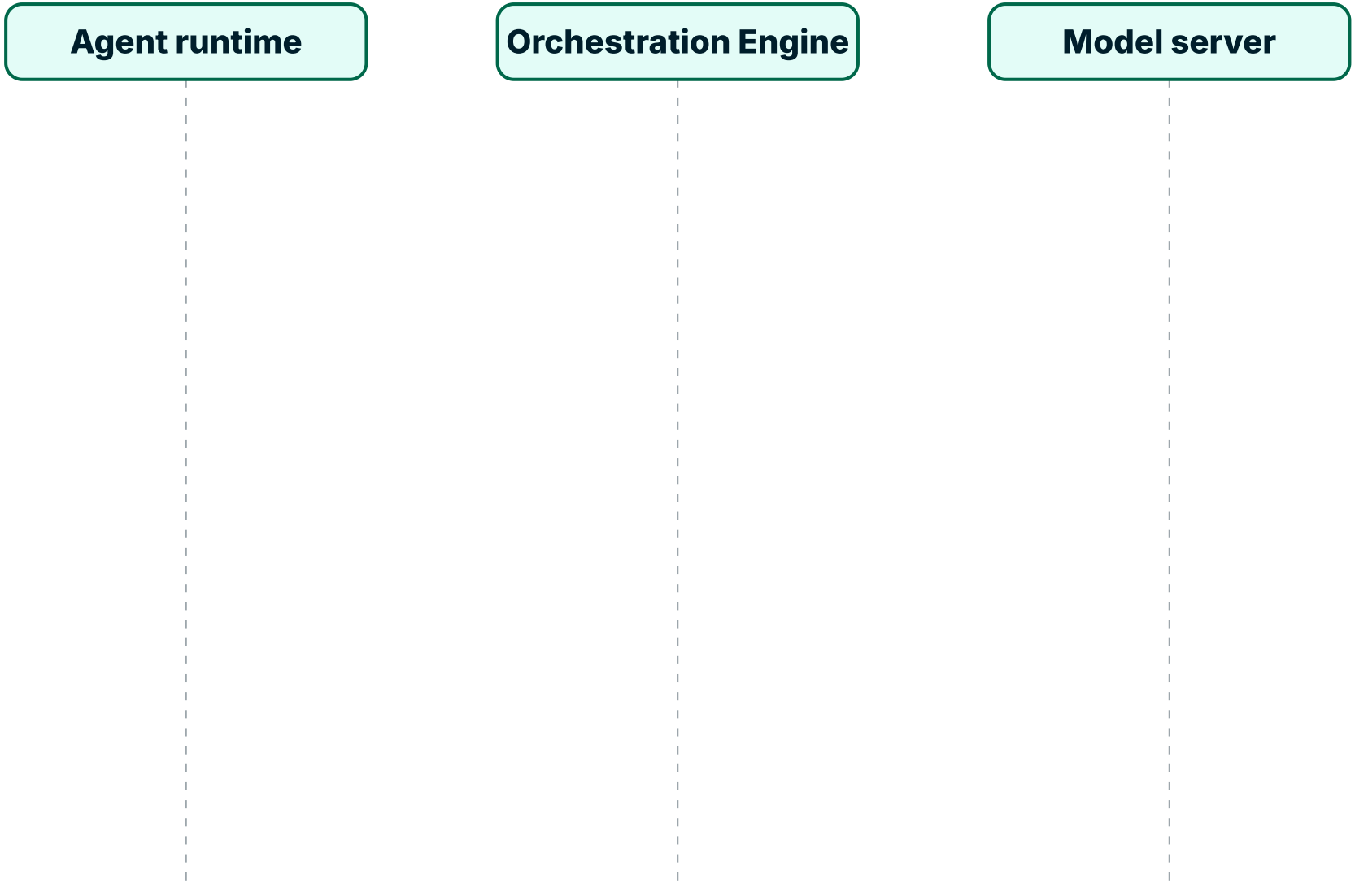
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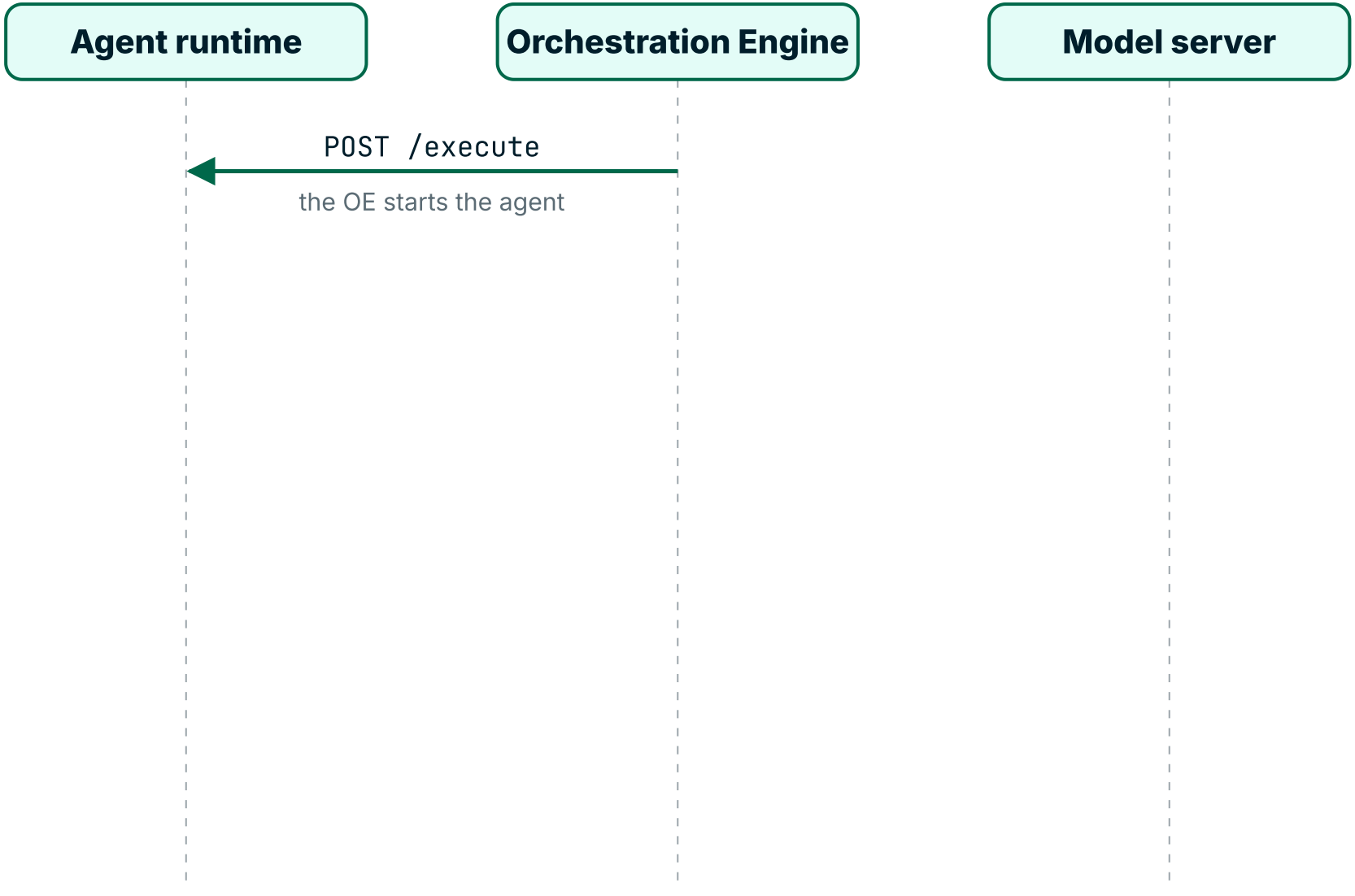
The app's log: *"durable Memory identity is incomplete"*

The template has memory on.
My request had no `user_id`. 🤦

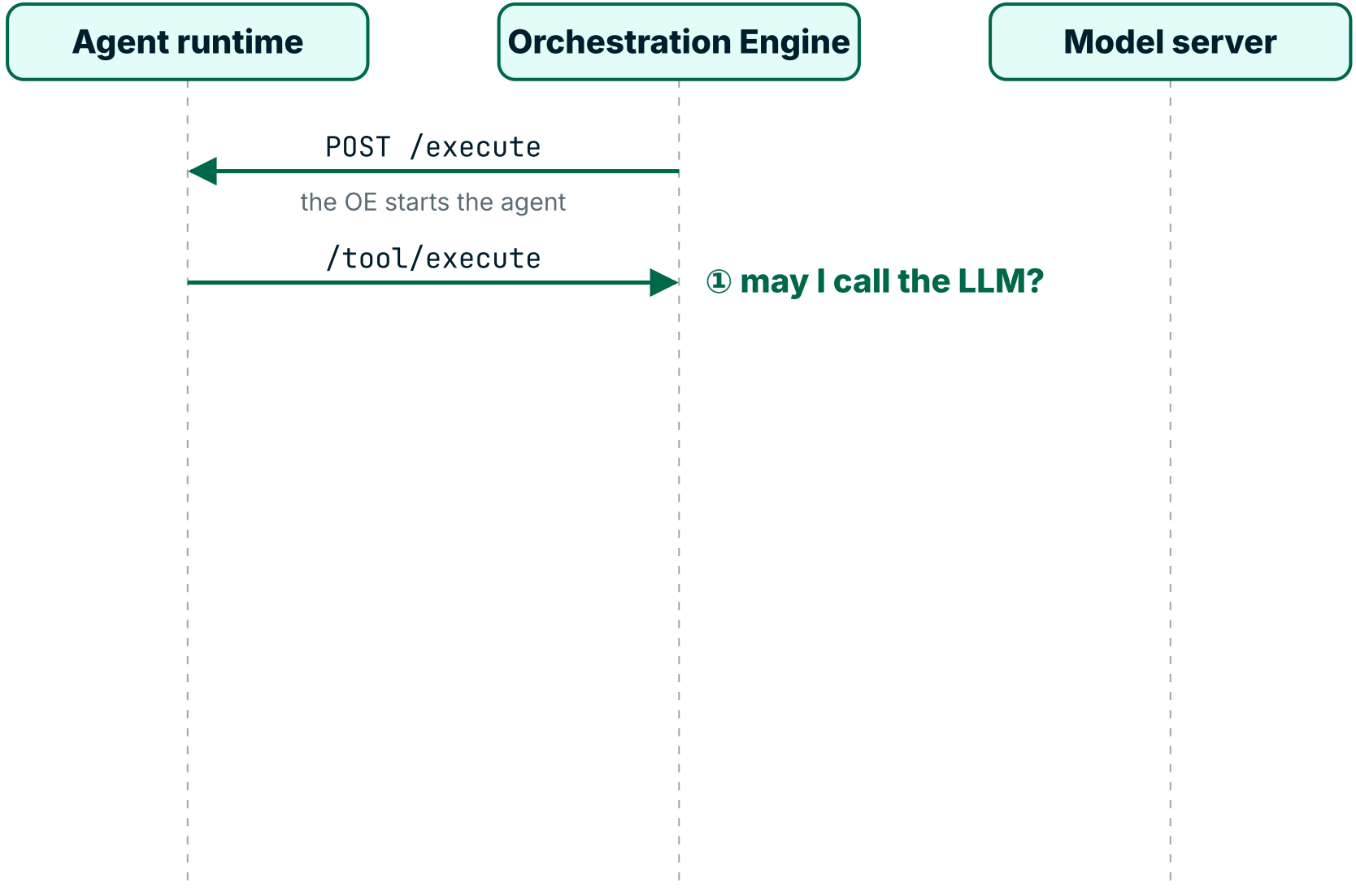
The runtime asks first



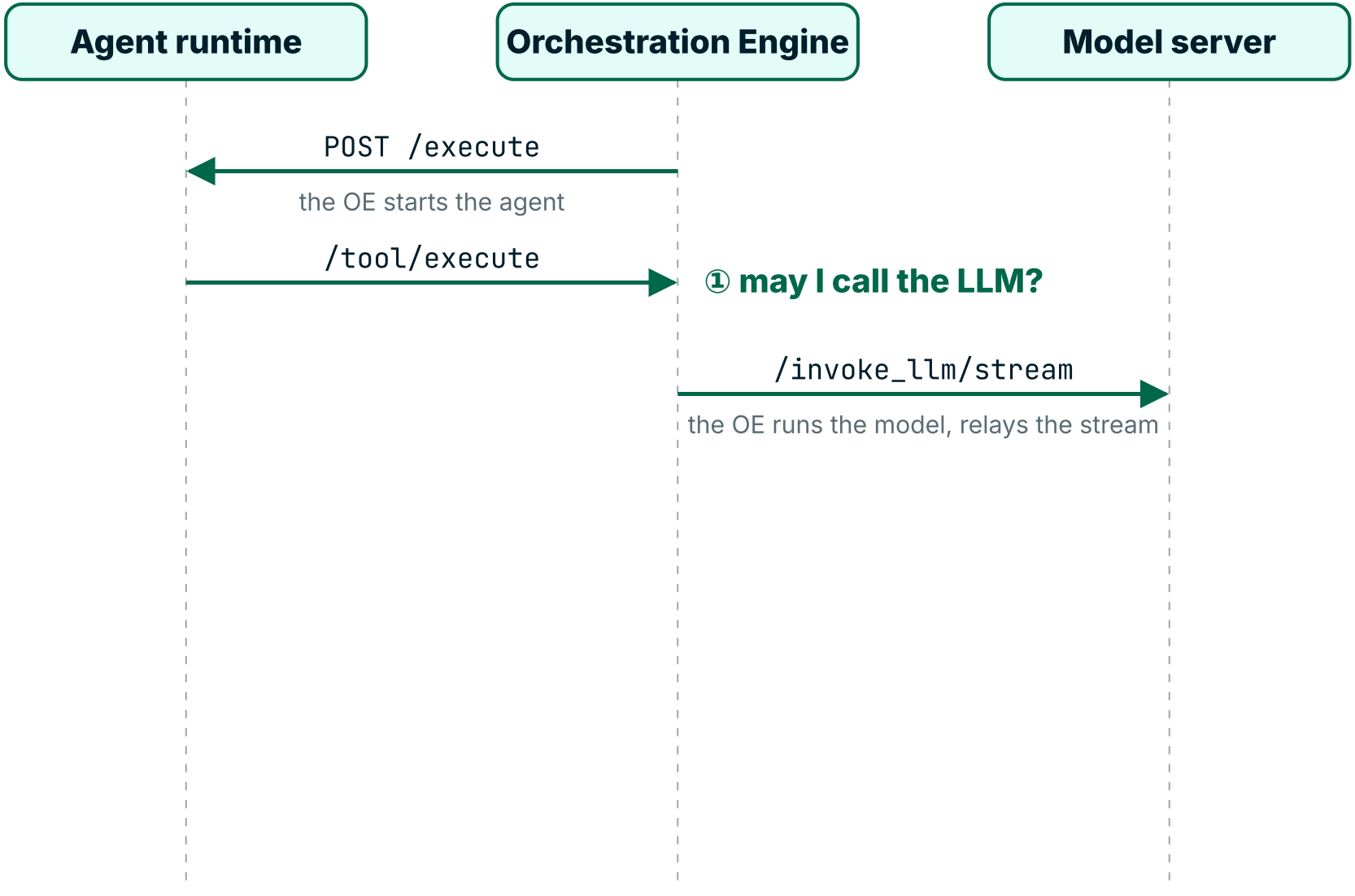
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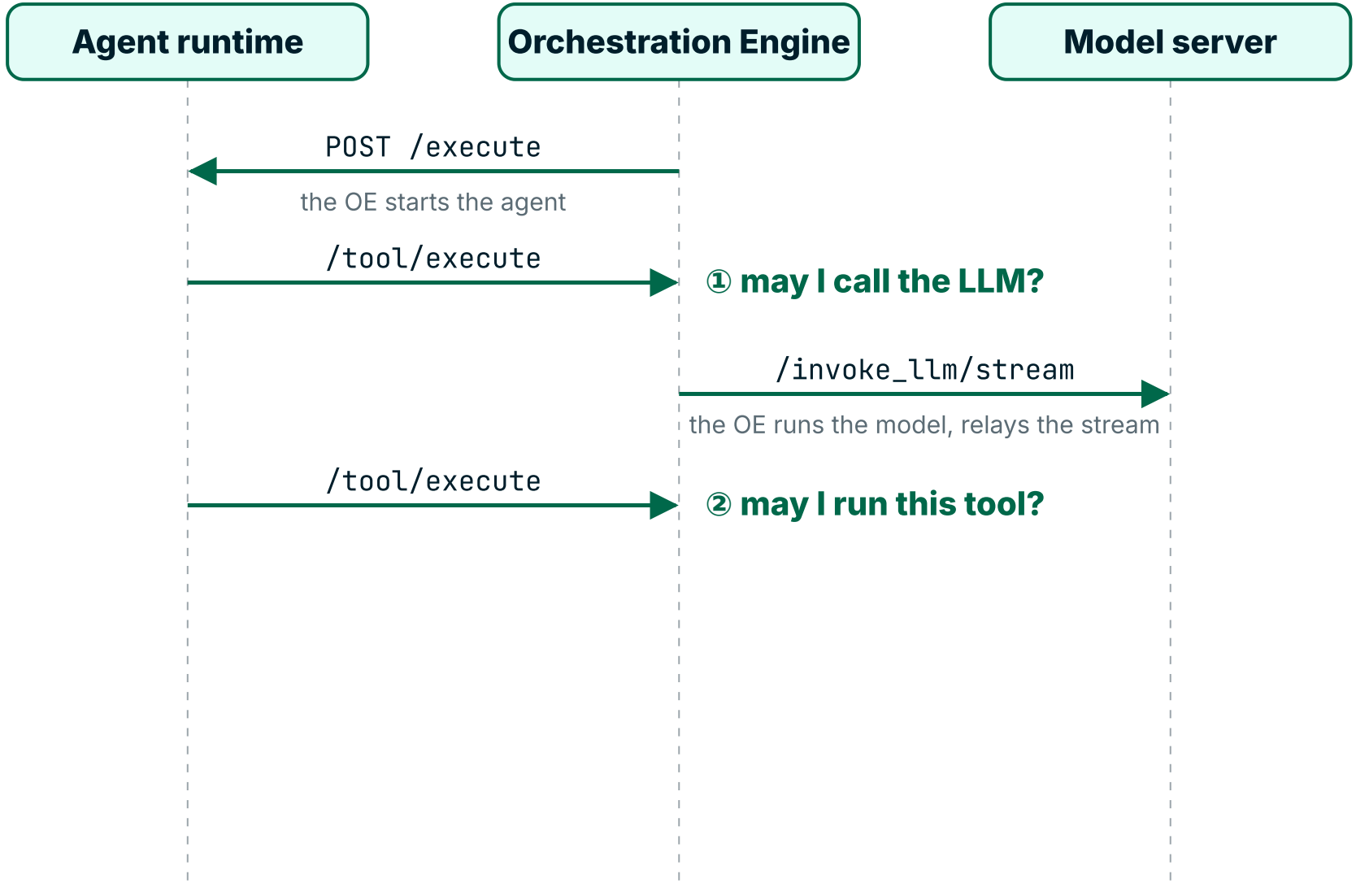
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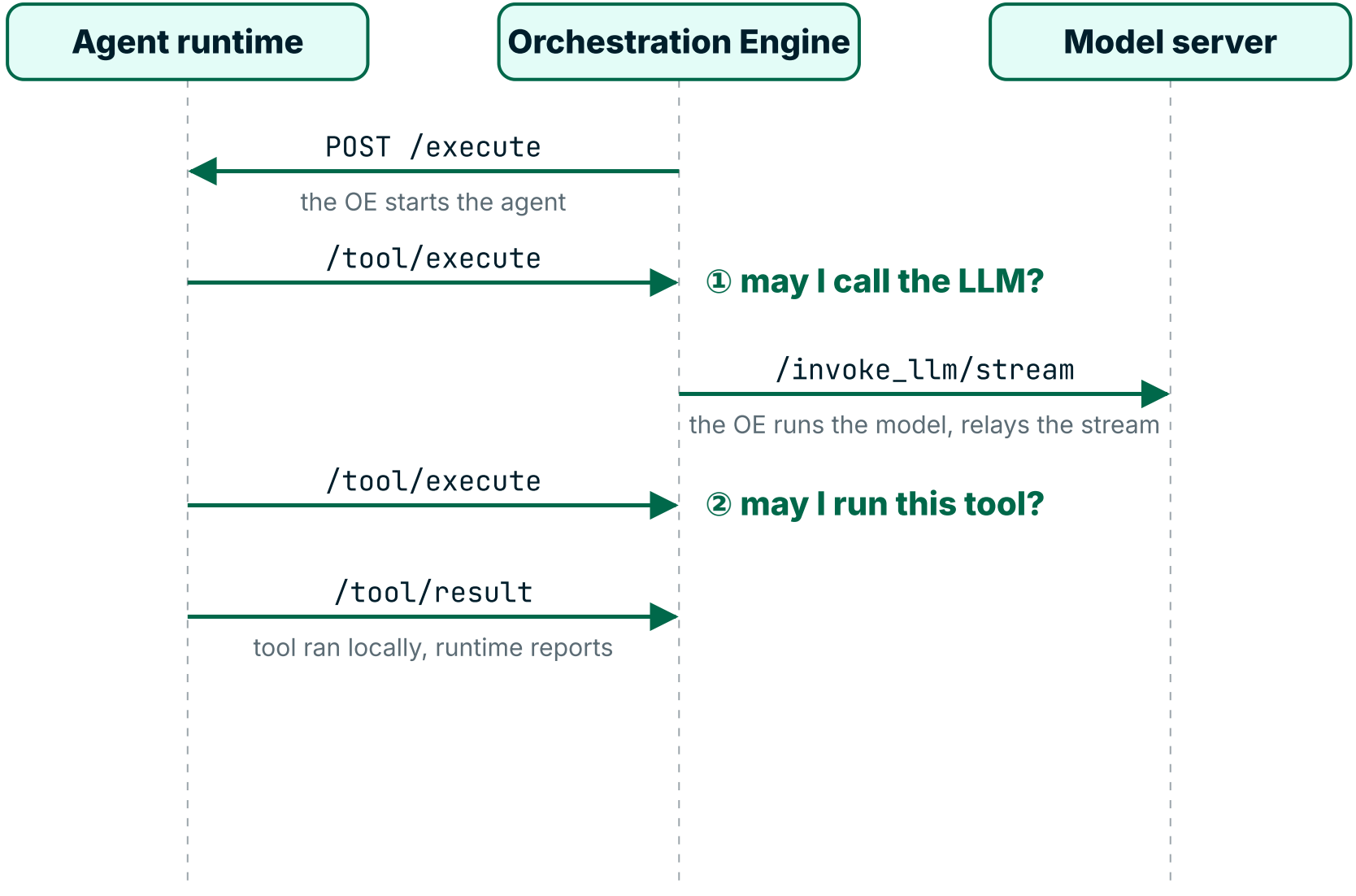
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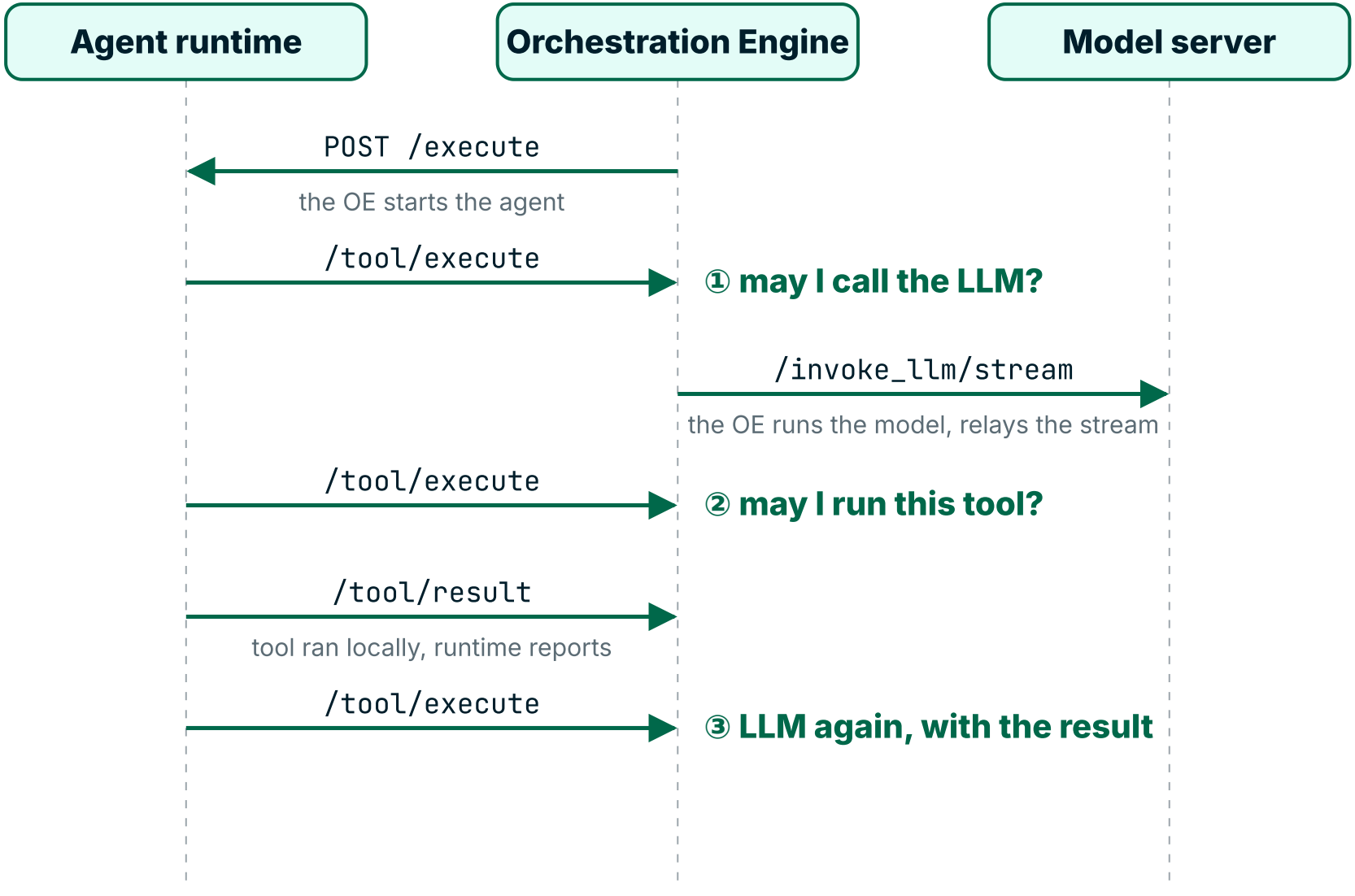
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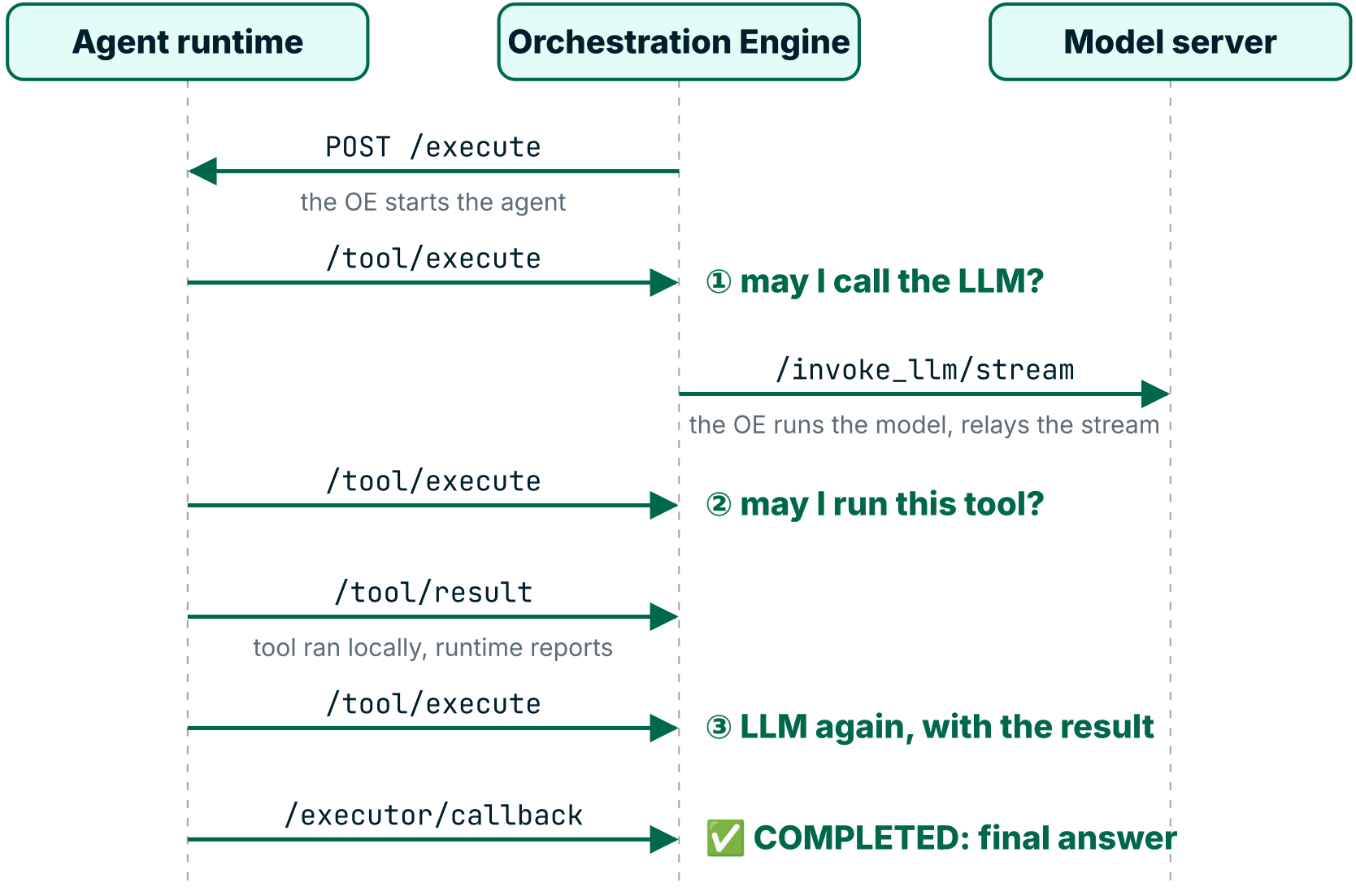
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Will the OE accept a Java container?

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```
{
  "result": "Hello ana, from a Java agent!",
  "status": "completed"
}
```

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`200` OK

```
{
  "result": "Hello ana, from a Java agent!",
  "status": "completed"
}
```

The OE didn't care that I'm Java. 🎉

CHAPTER 4

Graphs, Threads and Amnesia

An agent as a graph

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> *"nodes do the work, edges tell what to do next."*
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> *— LangGraph docs, Graph API › Graphs*

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1. 🧵 **Threads**: nodes run on LangGraph4j's own workers

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2. 💾 **Checkpoints**: the state saved after every node

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1. 🧵 **Threads**: nodes run on LangGraph4j's own workers
2. 💾 **Checkpoints**: the state saved after every node
3. ⏸️ **Pauses**: a human decides when the graph resumes

Problem #3: No execution on this thread

```
IllegalStateException: No execution is active on this thread
```

Whose thread is it?

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a thread-bound scope, so nothing gets passed around

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💥 **The worker**: "Execution? What execution?"

`IllegalStateException`

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🕸️ **LangGraph4j**: "Cool, I run every node on my own worker."
a different thread for each node

💥 **The worker**: "Execution? What execution?"
`IllegalStateException`

🧪 **A unit test**: "Caught it. Before the live agent did."
the graph test, not the demo

The fix

ScopedExecutor.java

```
// on the thread that has the execution
this.execution = ExecutionScope.current();

// on whichever worker picks the task up
POOL.execute(() ->
    ExecutionScope.call(execution, task));
```

The fix

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LangGraph4jAgent.java

```
 RunnableConfig.builder().threadId(threadId)
+    .emitterExecutor(new ScopedExecutor())
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Context tied to a thread stays on that thread.

Problem #4: The agent forgets

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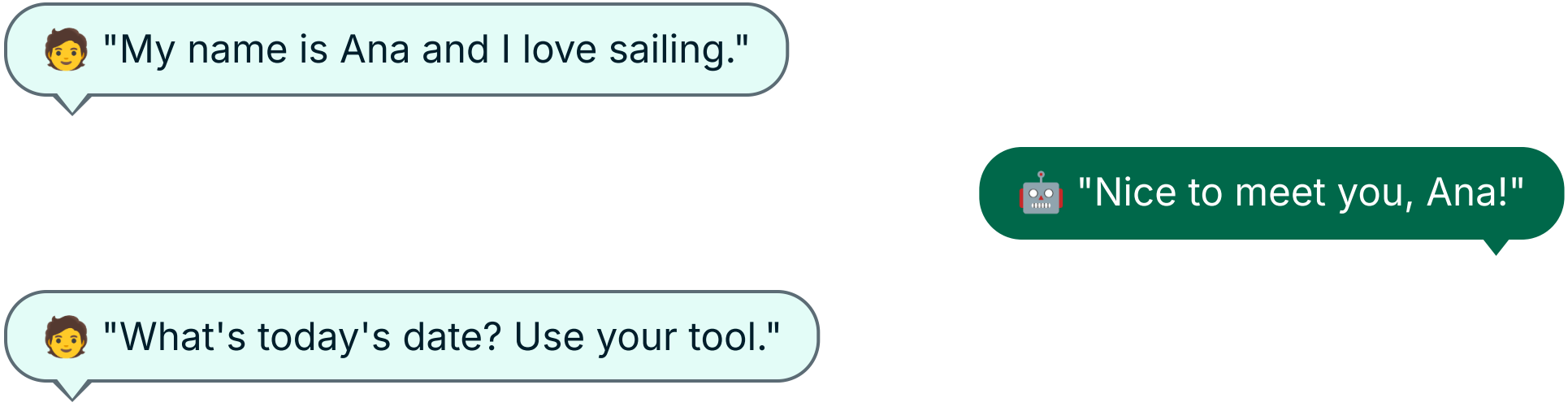
🧑 "My name is Ana and I love sailing."

Problem #4: The agent forgets

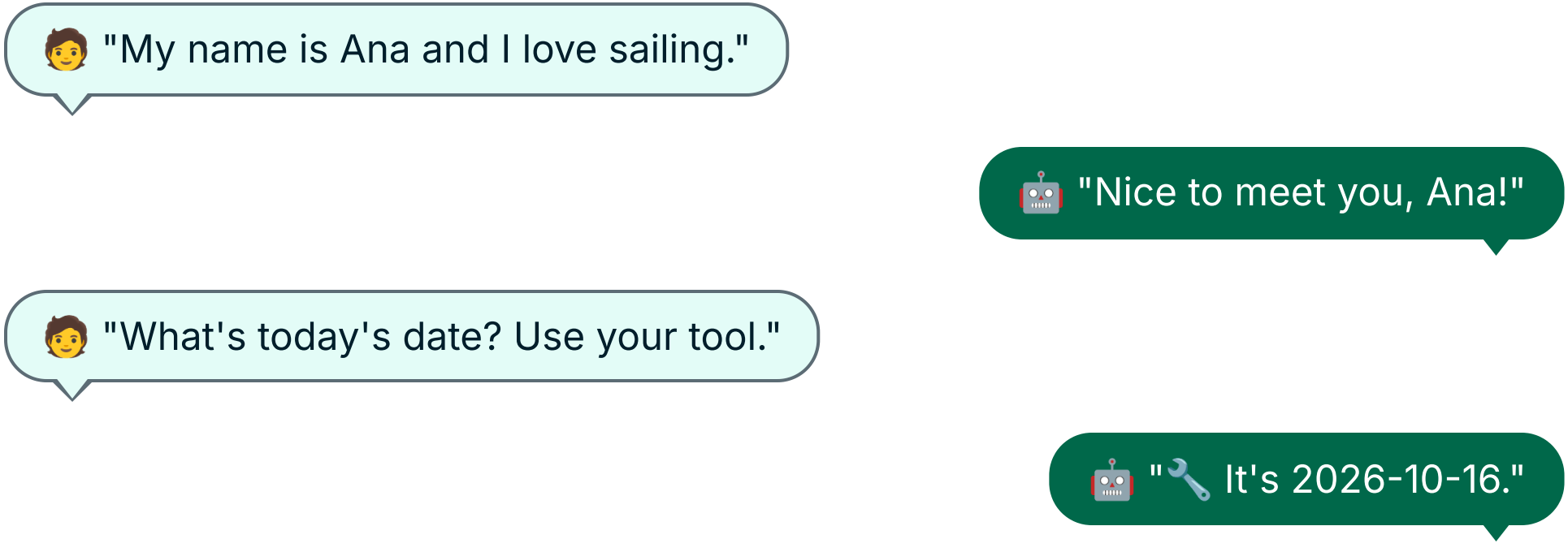
🧑 "My name is Ana and I love sailing."

🤖 "Nice to meet you, Ana!"

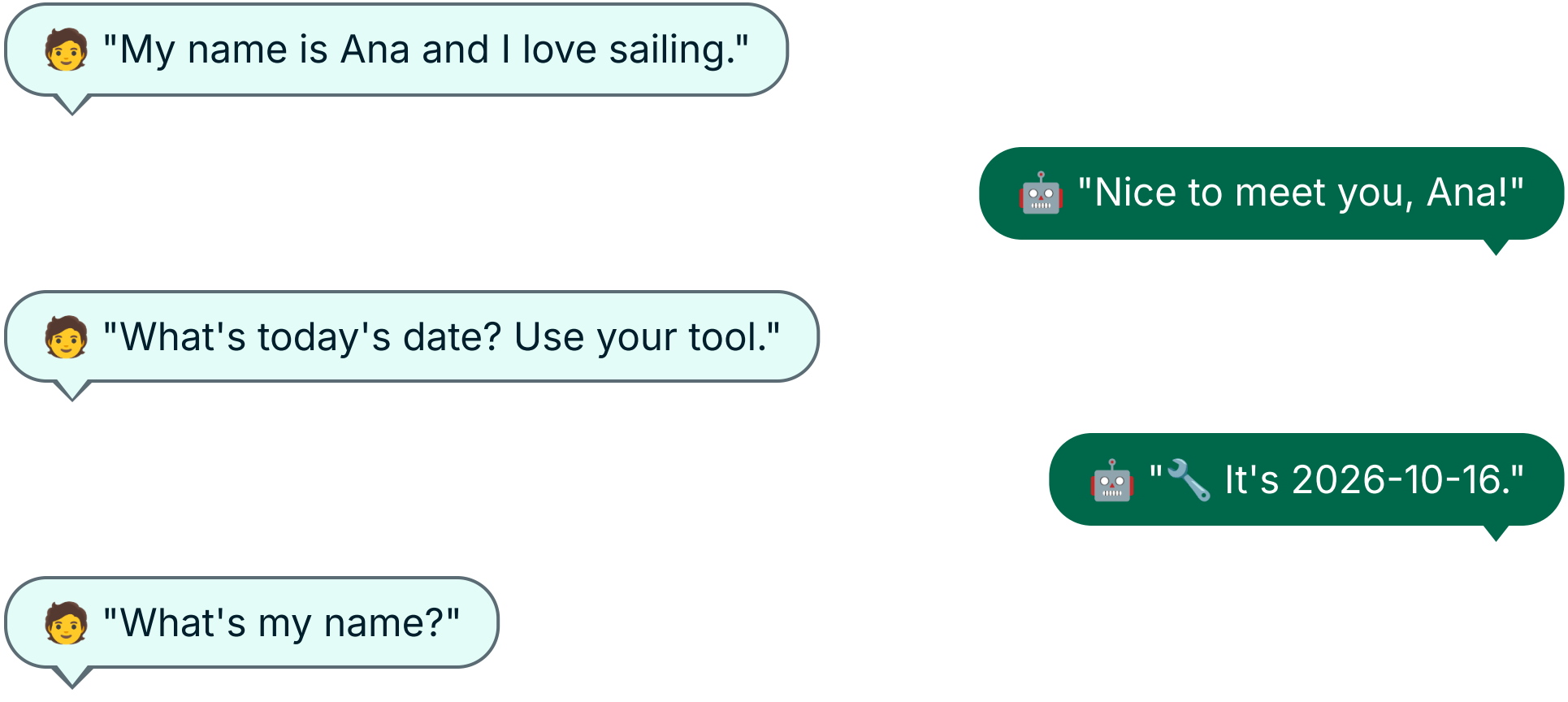
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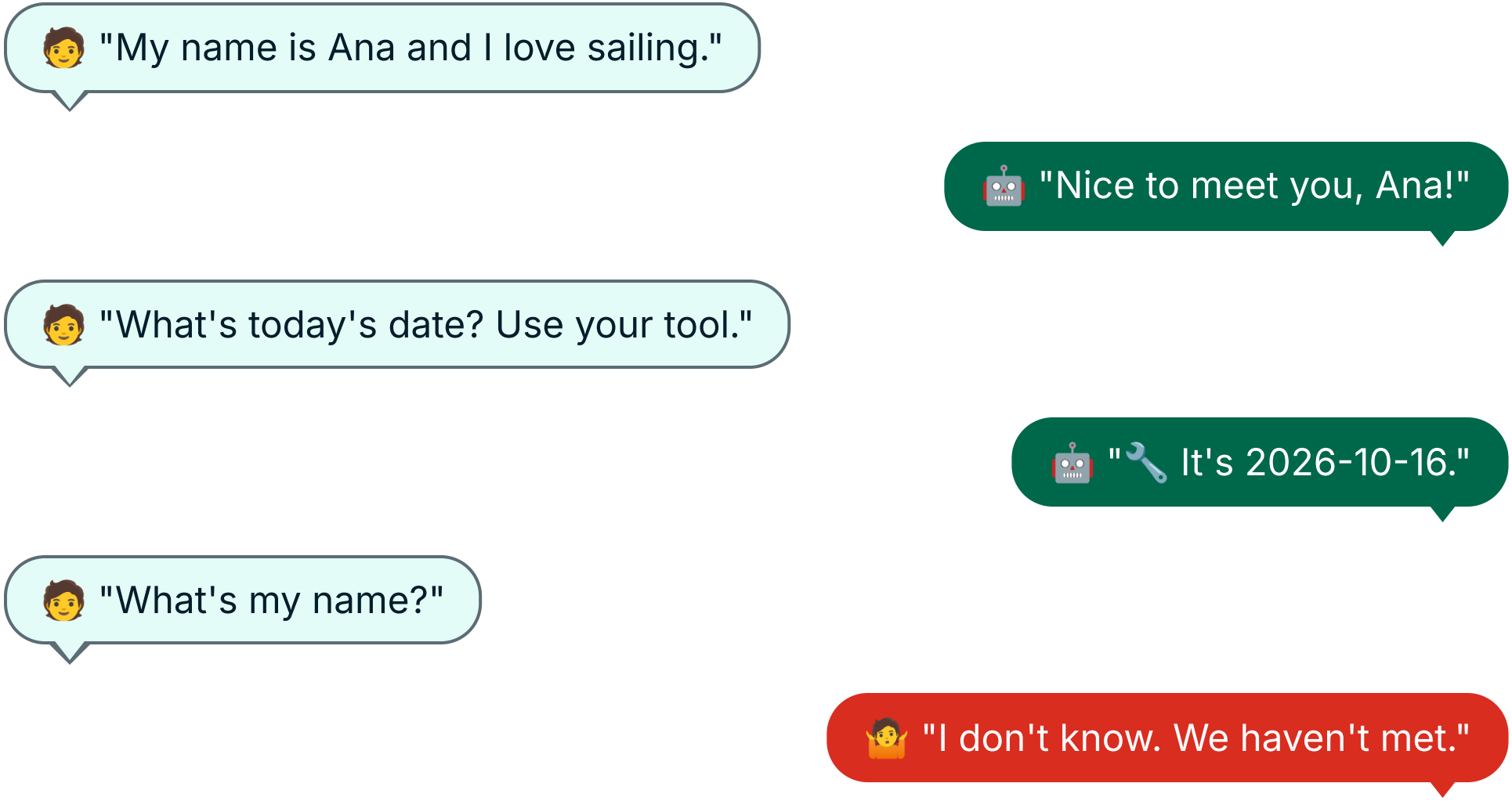
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nothing left to resume from

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nothing left to resume from

One agent turn is one run. A conversation is many.

The fix

App.java

```
 CompileConfig.builder()
     .checkpointSaver(checkpointer)
+    .releaseThread(false)   // keep it after the run
```

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// one session, one thread, every turn
threadId(session, workspaceId)
```

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⛵ "Your name is Ana, and you love sailing."

Memory that doesn't last ⚠️

Postgres

Redis

MySQL

Oracle

SQLite

DynamoDB

CockroachDB

Hazelcast

Memory that doesn't last ⚠️

Postgres | Redis | MySQL | Oracle | SQLite

DynamoDB | CockroachDB | Hazelcast | 🍃 MongoDB

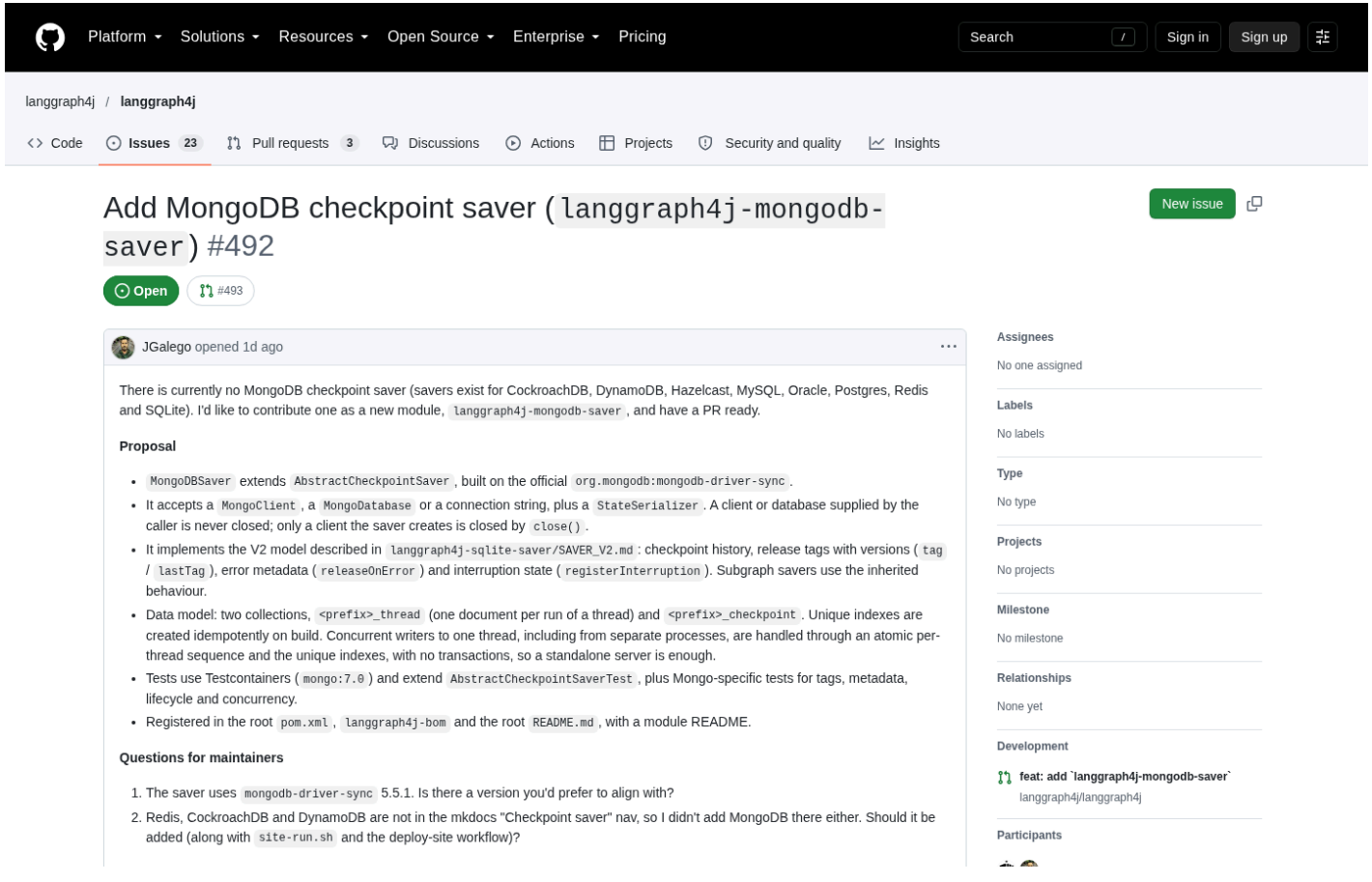
Memory that doesn't last ⚠️

Postgres | Redis | MySQL | Oracle | SQLite

DynamoDB | CockroachDB | Hazelcast | 🍃 MongoDB

Restart the runtime and the agent forgets again.

Issue #492: a MongoDB saver



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langgraph4j / langgraph4j

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Add MongoDB checkpoint saver (`langgraph4j-mongodb-saver`) #492

New issue

Open #493

JGalego opened 1d ago

There is currently no MongoDB checkpoint saver (savers exist for CockroachDB, DynamoDB, Hazelcast, MySQL, Oracle, Postgres, Redis and SQLite). I'd like to contribute one as a new module, `langgraph4j-mongodb-saver`, and have a PR ready.

Proposal

- `MongoDBSaver` extends `AbstractCheckpointSaver`, built on the official `org.mongodb:mongodb-driver-sync`.
- It accepts a `MongoClient`, a `MongoDatabase` or a connection string, plus a `StateSerializer`. A client or database supplied by the caller is never closed; only a client the saver creates is closed by `close()`.
- It implements the V2 model described in `langgraph4j-sqlite-saver/SAVER_V2.md`: checkpoint history, release tags with versions (`tag` / `lastTag`), error metadata (`releaseOnError`) and interruption state (`registerInterruption`). Subgraph savers use the inherited behaviour.
- Data model: two collections, `<prefix>_thread` (one document per run of a thread) and `<prefix>_checkpoint`. Unique indexes are created idempotently on build. Concurrent writers to one thread, including from separate processes, are handled through an atomic per-thread sequence and the unique indexes, with no transactions, so a standalone server is enough.
- Tests use Testcontainers (`mongo:7.0`) and extend `AbstractCheckpointSaverTest`, plus Mongo-specific tests for tags, metadata, lifecycle and concurrency.
- Registered in the root `pom.xml`, `langgraph4j-bom` and the root `README.md`, with a module README.

Questions for maintainers

1. The saver uses `mongodb-driver-sync` 5.5.1. Is there a version you'd prefer to align with?
2. Redis, CockroachDB and DynamoDB are not in the mkdocs "Checkpoint saver" nav, so I didn't add MongoDB there either. Should it be added (along with `site-run.sh` and the deploy-site workflow)?

Assignees
No one assigned

Labels
No labels

Type
No type

Projects
No projects

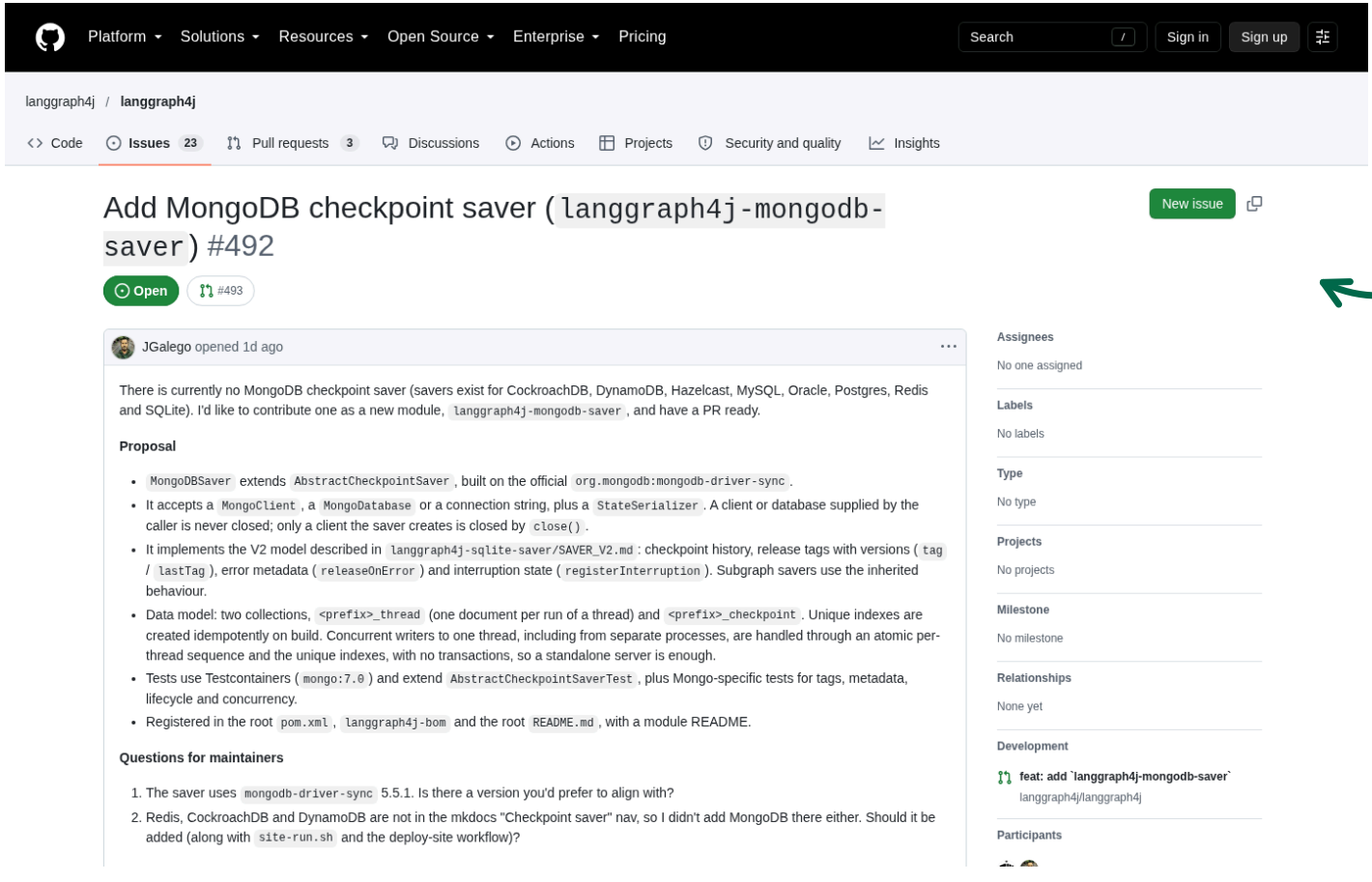
Milestone
No milestone

Relationships
None yet

Development
feat: add `langgraph4j-mongodb-saver`
langgraph4j/langgraph4j

Participants

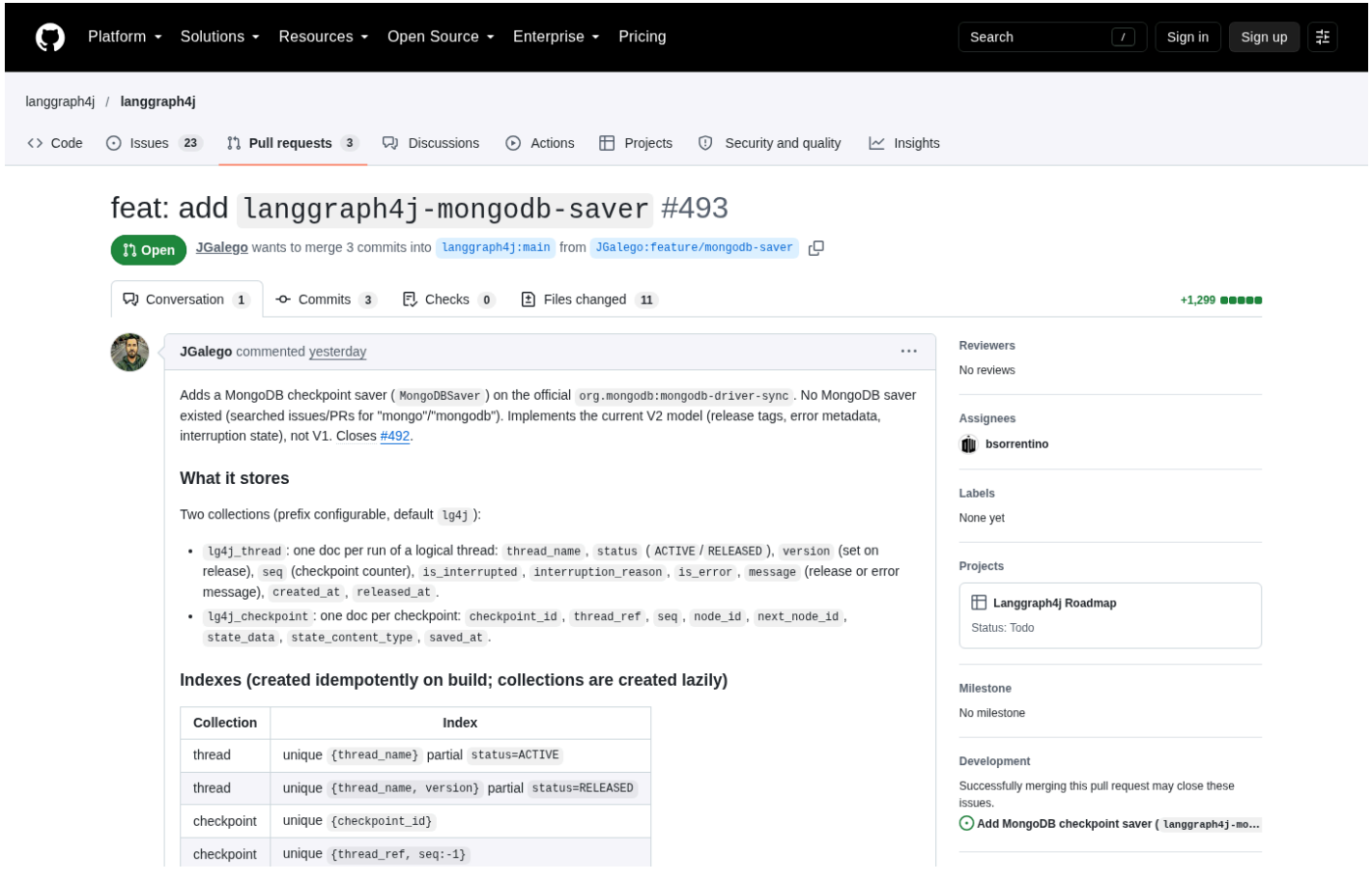
Issue #492: a MongoDB saver



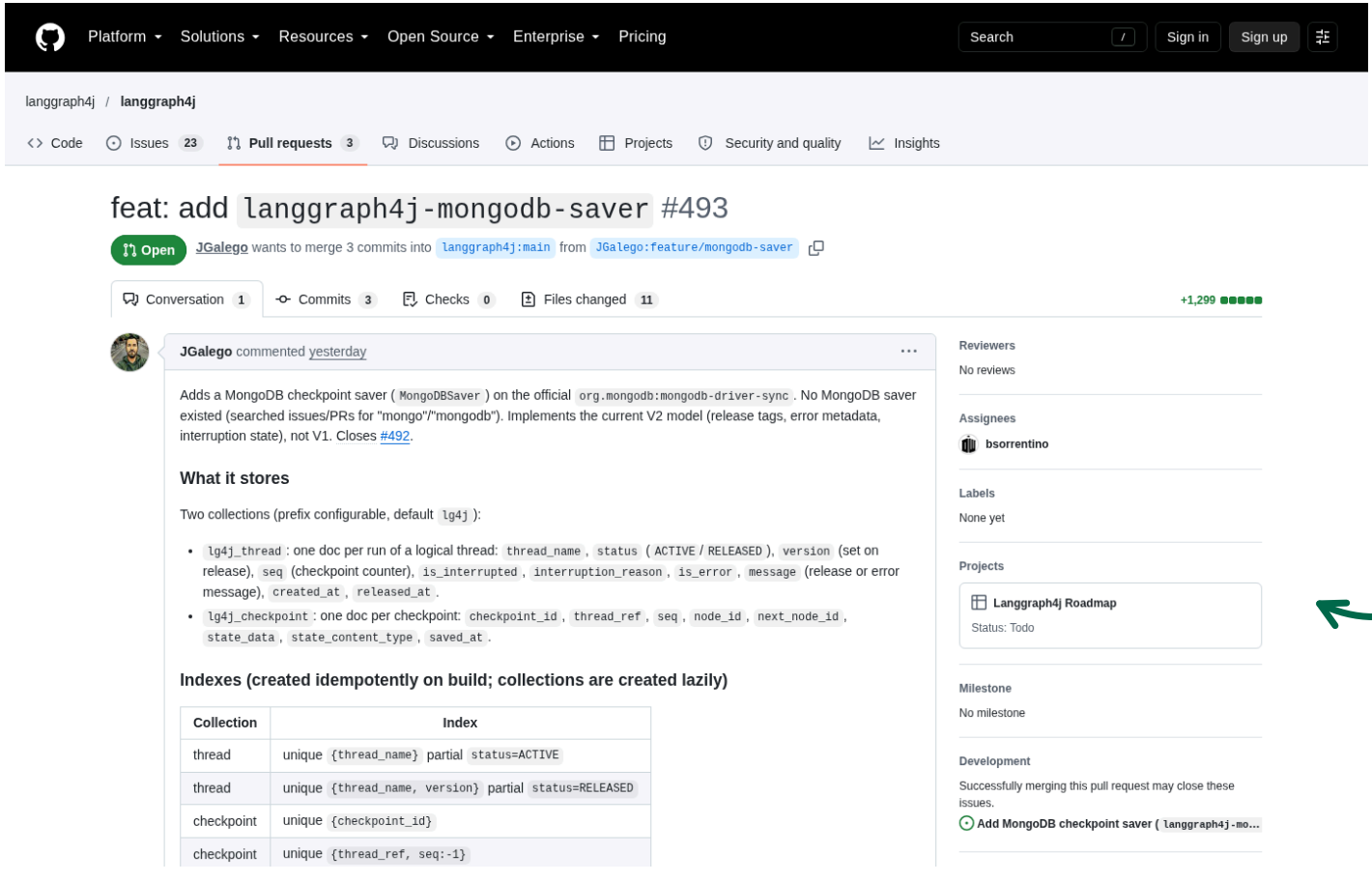
open!

Oct 15

PR #493: the saver

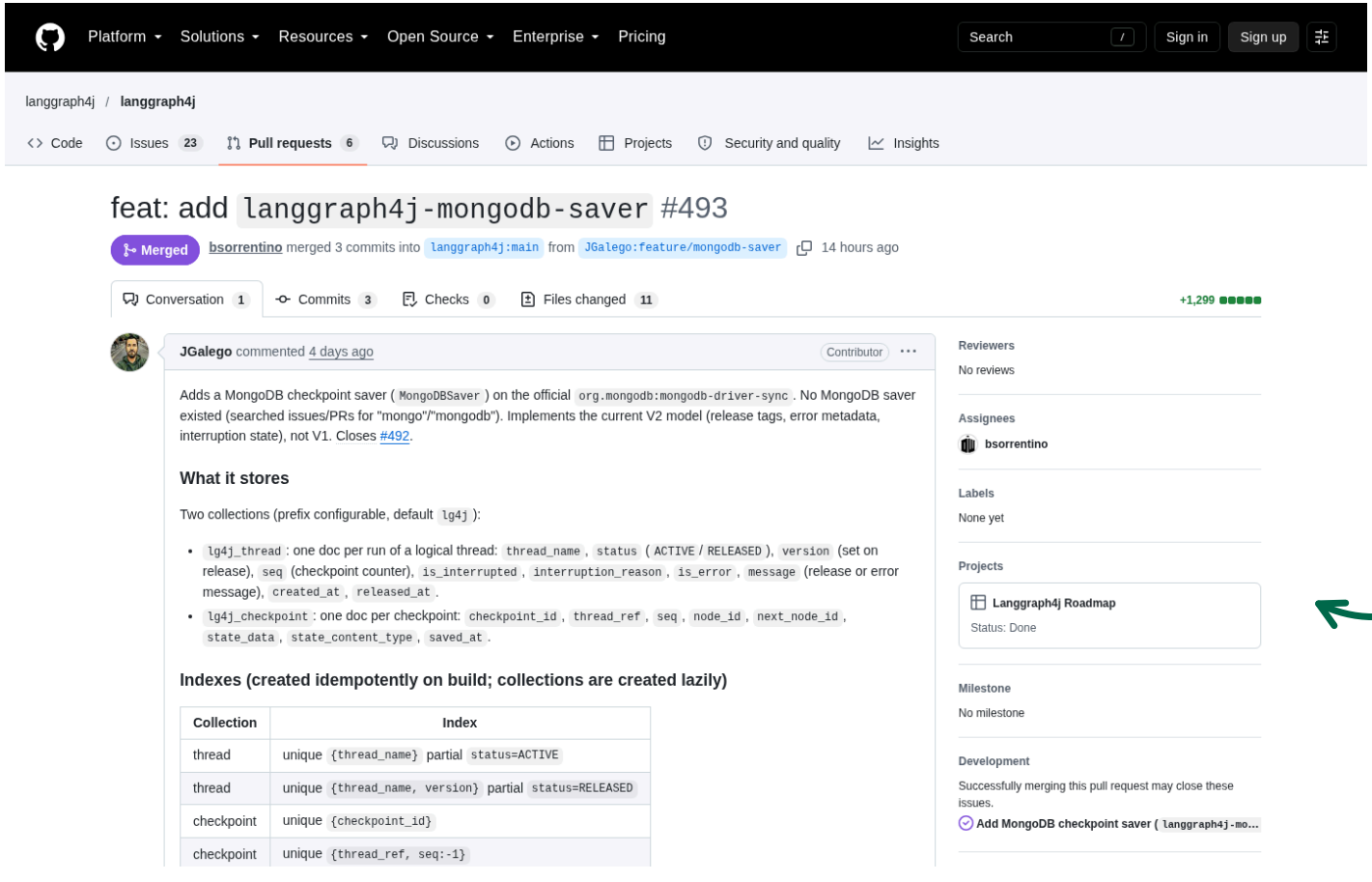


PR #493: the saver



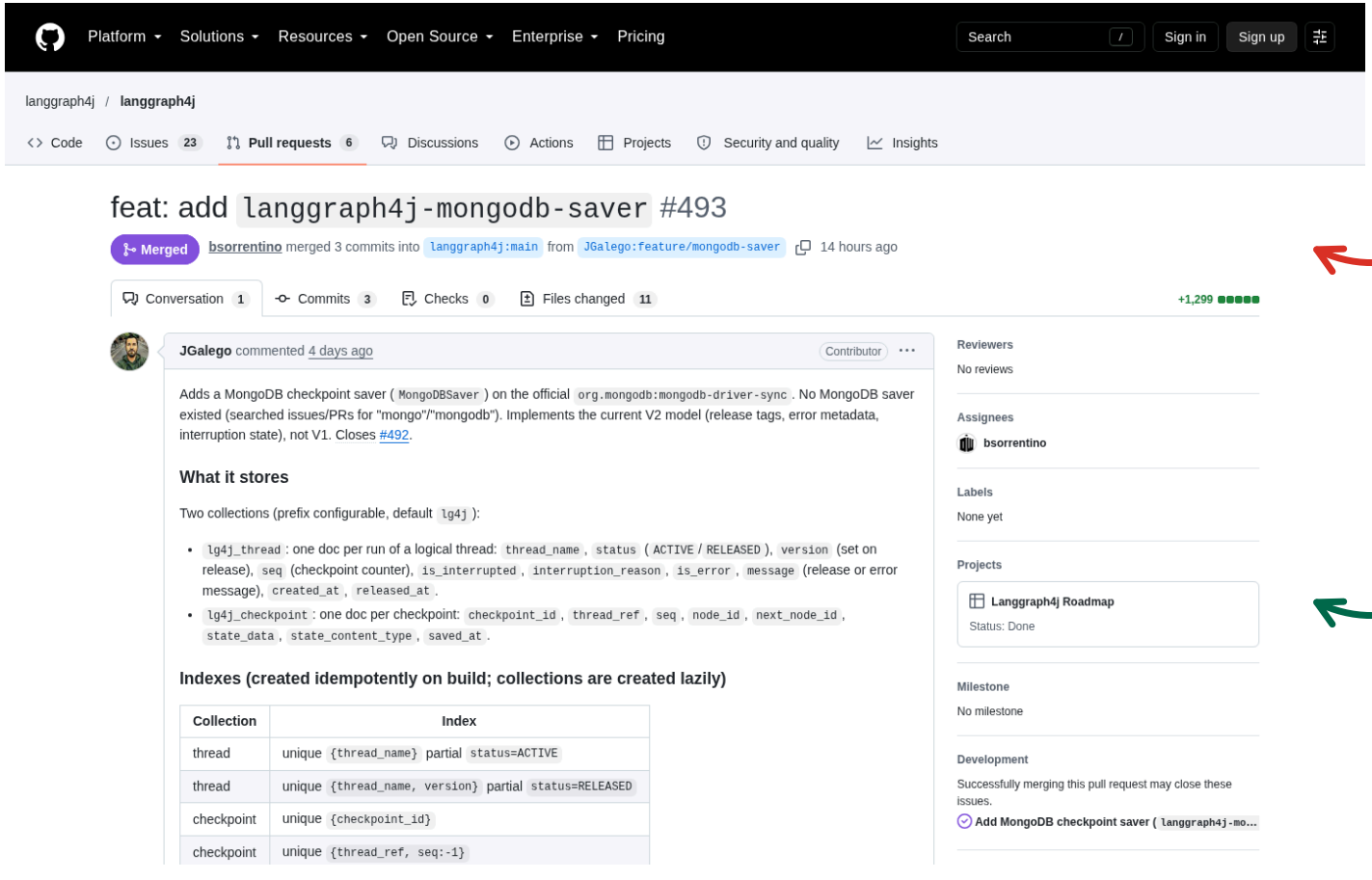
on the roadmap!

PR #493: the saver



on the roadmap!

PR #493: the saver



merged!





on the roadmap!



Problem #5: The resume decision is nested

```
throw new SuspendException("awaiting_human_review", context);
```

Problem #5: The resume decision is nested

```
throw new SuspendException("awaiting_human_review", context);
```

What does the agent receive when it comes back? 🤔

What did the OE actually send?

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🤖 **Claude**: "Easy. `resumeData.decision`."
guess #1, from the docs. Read an empty string.

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🧠 **The OE**: "I nested it: `human_review`, then `decision`."

not in the docs

What did the OE actually send?

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🖨️ **The demo agent**: "Here's what arrived."
it printed the payload. Mystery solved.

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not in the docs

🖨️ **The demo agent**: "Here's what arrived."
it printed the payload. Mystery solved.

When the docs are silent, print what arrived.

The fix

ApprovalAgent.java

```
-ctx.resumeData().path("decision")
+ctx.resumeData().path("human_review").path("decision")
```

The fix

📄 ApprovalAgent.java

```
-ctx.resumeData().path("decision")
+ctx.resumeData().path("human_review").path("decision")
```

🧑 The reviewer's answer

```
POST /resume/<execution-id>
{"human_review": {"decision": "approve"}}
```

The fix

📄 ApprovalAgent.java

```
-ctx.resumeData().path("decision")
+ctx.resumeData().path("human_review").path("decision")
```

🧑 The reviewer's answer

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POST /resume/<execution-id>
{"human_review": {"decision": "approve"}}
```

👍 approve → "Approved, going ahead."

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-ctx.resumeData().path("decision")
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```

🧑 The reviewer's answer

```
POST /resume/<execution-id>
{"human_review": {"decision": "approve"}}
```

👍 approve → "Approved, going ahead."

👎 reject → "Not approved, nothing was done."

CHAPTER 5

Reports, Not Findings

Problem #6: Every pump query returns boilerplate

```
> insulin pump delivered too much insulin

1. "insufficient information regarding a device problem…"
```

Why?

Why?

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so long narratives get split into chunks

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🏥 **A MAUDE narrative**: "...and then the manufacturer's boilerplate."

the later chunks say nothing about the event

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🧮 **MiniLM**: "I read 256 wordpieces. The rest is cut."
so long narratives get split into chunks

🏥 **A MAUDE narrative**: "...and then the manufacturer's boilerplate."
the later chunks say nothing about the event

🔎 **The index**: "Boilerplate is close to everything."
so it ranked first for every query

Why?

🧮 **MiniLM**: "I read 256 wordpieces. The rest is cut."
so long narratives get split into chunks

🏥 **A MAUDE narrative**: "...and then the manufacturer's boilerplate."
the later chunks say nothing about the event

🔎 **The index**: "Boilerplate is close to everything."
so it ranked first for every query

Generic text is close to every question.

The fix

MaudeIngestor.java

```
// 120 words a chunk, 30 shared, and only the first chunk
TextChunks.split(narrative, 120, 30, 1)
```

The fix

MaudeIngestor.java

```
// 120 words a chunk, 30 shared, and only the first chunk
TextChunks.split(narrative, 120, 30, 1)
```

Keep the part where someone says what happened.

Reports, not findings

NearMissAgent.java · the system prompt

> The reports are submissions, not findings. Never say that a device, a vehicle or an aircraft is defective, that it caused an outcome, or what a patient, driver or pilot should do. Point to the report instead.

A report is a claim

A report is a claim

🏥 **A report**: "The pump was confirmed defective by medical staff."

what somebody submitted

A report is a claim

🏥 **A report**: "The pump was confirmed defective by medical staff."

what somebody submitted

🤖 **The agent**: "One report states that the pump was defective."

attributed, with the report number

A report is a claim

🏥 **A report**: "The pump was confirmed defective by medical staff."
what somebody submitted

🤖 **The agent**: "One report states that the pump was defective."
attributed, with the report number

🚫 **Never**: "The pump is defective."
nobody confirmed it

When is a connection real?

📄 same report

🏷️ same product code

🚙 same vehicle

🛩️ same aircraft

📚 same report set

When is a connection real?

📄 same report

🏷️ same product code

🚙 same vehicle

✈️ same aircraft

📚 same report set

`correlate_evidence` sets `verified_connection: true` only when two records share one of these.

A pacemaker and a Tesla

A pacemaker and a Tesla

🧑 "Is there a verified connection between the pacemaker reports and the Tesla Model 3 complaints about sudden loss of power?"

A pacemaker and a Tesla

🧑 "Is there a verified connection between the pacemaker reports and the Tesla Model 3 complaints about sudden loss of power?"

verified_same_* shared report, product code, vehicle, aircraft...

analogous_pattern similar vectors, collected independently

independent similarity below the reporting cutoff

What the tool said

What the tool said

🔧 **correlate_evidence**
MAUDE record vs NHTSA record

What the tool said

🔧 **correlate_evidence**
MAUDE record vs NHTSA record


```
{
  "left_dataset_id": "fda-maude",
  "right_dataset_id": "nhtsa-complaints",
  "relationship": "independent",
  "verified_connection": false,
  "explanation": "…The datasets were collected independently.
                  This is not a verified connection."
}
```

What the tool said

🔧 **correlate_evidence**
MAUDE record vs NHTSA record


```
{
  "left_dataset_id": "fda-maude",
  "right_dataset_id": "nhtsa-complaints",
  "relationship": "independent",
  "verified_connection": false,
  "explanation": "…The datasets were collected independently.
                  This is not a verified connection."
}
```


🤖 "There is no verified connection."

CHAPTER 6

The Stowaway Agent

The wall

```
$ agentengine init

Error: unsupported language/framework combination "java-none"
```

The wall

```
$ agentengine init

Error: unsupported language/framework combination "java-none"
```

☝️ **Me**: "A hidden option: `AGENTENGINE_IMAGE_SOURCE=custom`. Can I bring my own image?"

The wall

```
$ agentengine init

Error: unsupported language/framework combination "java-none"
```

☝️ **Me**: "A hidden option: `AGENTENGINE_IMAGE_SOURCE=custom`. Can I bring my own image?"

🔎 **The template**: "I only mirror the CLI binary and the dev-stack images."
from `agent source template --stdout`

No Java, no Maven

🐍 `pyproject.toml` → `runner-base-python-lite`

No Java, no Maven

🐍 pyproject.toml → runner-base-python-lite

🟨 package.json → runner-base-typescript-langgraph

No Java, no Maven



The loophole

```
app = _load_app(config["entrypoint"])   # must be a BaseApp
app.run()
```

The loophole

🏗️ **The platform**: "I import whatever `agent.yaml` names, check that it's a `BaseApp`, and call `run()`."

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The loophole

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```
app = _load_app(config["entrypoint"])   # must be a BaseApp
app.run()
```

Nobody said `run()` has to return.

The stowaway

java_shim/__init__.py

```
class JavaApp(BaseApp):
    def run(self):
        from jdk4py import JAVA           # a JDK... from pip
        os.execv(str(JAVA), [str(JAVA), "-cp", "java/lib/*",
                 "com.mongodb.agentengine.runner.Launcher"])

app = JavaApp("java-agent")
```

The stowaway

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class JavaApp(BaseApp):
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1. 🐍 A Python package on paper

The stowaway

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The stowaway

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app = JavaApp("java-agent")
```

1. 🐍 A Python package on paper
2. ☕ A JRE from PyPI inside a wheel
3. 📦 Prebuilt jars in the source tree

Will the build buy it?

```
$ agentengine build logs bld_01M524H4BZG1YJJJQRGJ5NZ6JM

2026-10-16T10:36:50Z Detected agent language: python

2026-10-16T10:37:02Z Using entrypoint: java_shim:app
```

Will the build buy it?

```
$ agentengine build logs bld_01M524H4BZG1YJJJQRGJ5NZ6JM

2026-10-16T10:36:50Z Detected agent language: python

2026-10-16T10:37:02Z Using entrypoint: java_shim:app
```

🏗️ **The platform**: "Ah yes, a Python agent." 🐍

Problem #7: The agent never starts

```
$ agentengine invoke "Hello from the managed platform"

    The agent did not become ready during startup.

$ agentengine logs --source agent

    SSLHandshakeException: PKIX path building failed
```

Who's to blame?

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🤖 **Claude**: "A private CA. Python trusts it, Java doesn't."

so I wrote a launcher that merged the bundles

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after the redeploy the handshake still failed

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mutual TLS: the client has to prove who it is, not only the server

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🧠 **The OE**: "I want a client certificate."
mutual TLS: the client has to prove who it is, not only the server

🐍 **The Python runner**: "`TLS_CERT_PEM`, `TLS_KEY_PEM`, `TLS_CA_CERT_PEM`. If any of the three is missing I refuse to connect."
the contract was written down only in **`tls_client.py`**

The fix

PlatformTls.java

```
// trust only the platform's CA
trust.setCertificateEntry("ca-" + index++, anchor);

// present the platform's client certificate and key
identity.setKeyEntry("client", privateKey, noPassword, chain);

context.init(keyManagers, trustManagers, null);
```

The fix

PlatformTls.java

```
// trust only the platform's CA
trust.setCertificateEntry("ca-" + index++, anchor);

// present the platform's client certificate and key
identity.setKeyEntry("client", privateKey, noPassword, chain);

context.init(keyManagers, trustManagers, null);
```

Built from the three TLS_* variables.
No certificate, no connection.

One more catch: the key

PlatformTls.java

```
case "RSA PRIVATE KEY" ->
    der(0x30, concat(INTEGER_ZERO, RSA_ALGORITHM, der(0x04, der)));
```

One more catch: the key

☕ **Java**: "I only read keys that start with `BEGIN PRIVATE KEY`."
the PKCS#8 format

📄 `PlatformTls.java`

```
case "RSA PRIVATE KEY" ->
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☕ **Java**: "I only read keys that start with `BEGIN PRIVATE KEY`."
the PKCS#8 format

🔑 **The platform**: "Mine may start with `BEGIN RSA PRIVATE KEY` or `BEGIN EC PRIVATE KEY`."
the older PKCS#1 and SEC1 formats

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```
case "RSA PRIVATE KEY" ->
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📄 `PlatformTls.java`

```
case "RSA PRIVATE KEY" ->
    der(0x30, concat(INTEGER_ZERO, RSA_ALGORITHM, der(0x04, der)));
```

The older keys get wrapped in a PKCS#8 envelope by hand.

Problem #8: The package is too big

```
object exceeds 52428800 byte limit
```



ONNX Runtime jar

130 MB · all OS natives

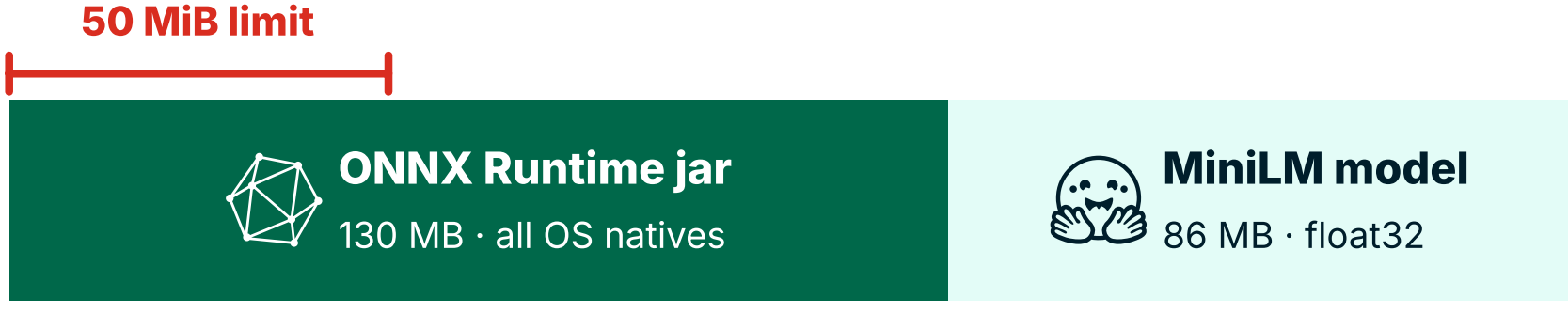


MiniLM model

86 MB · float32

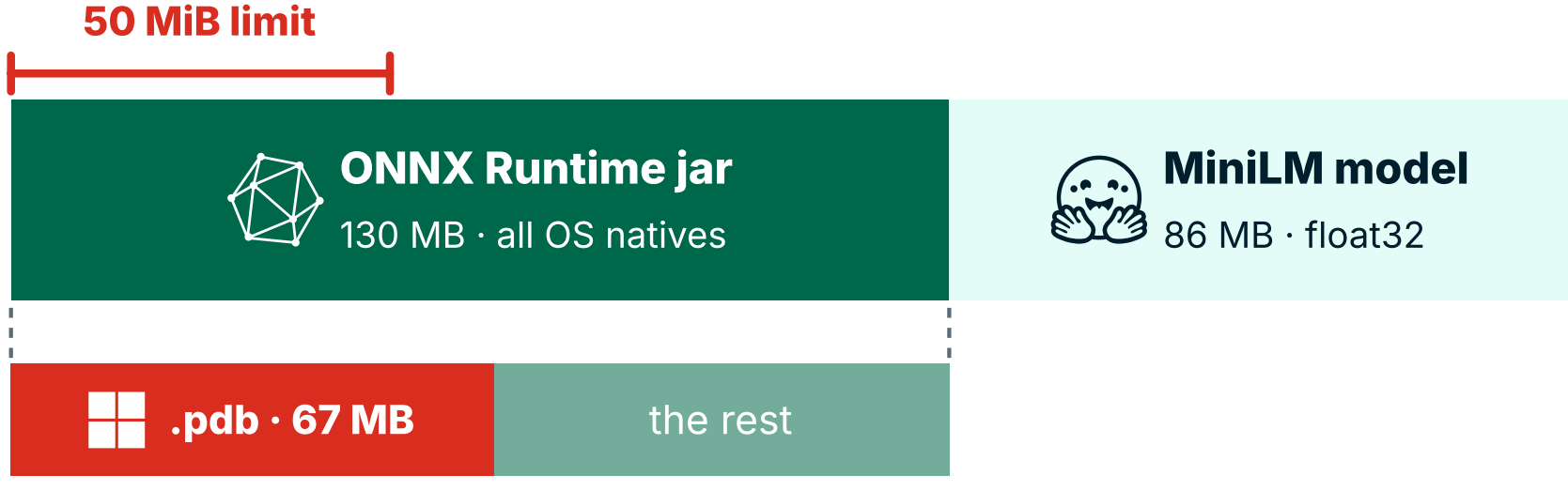
Problem #8: The package is too big

```
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```

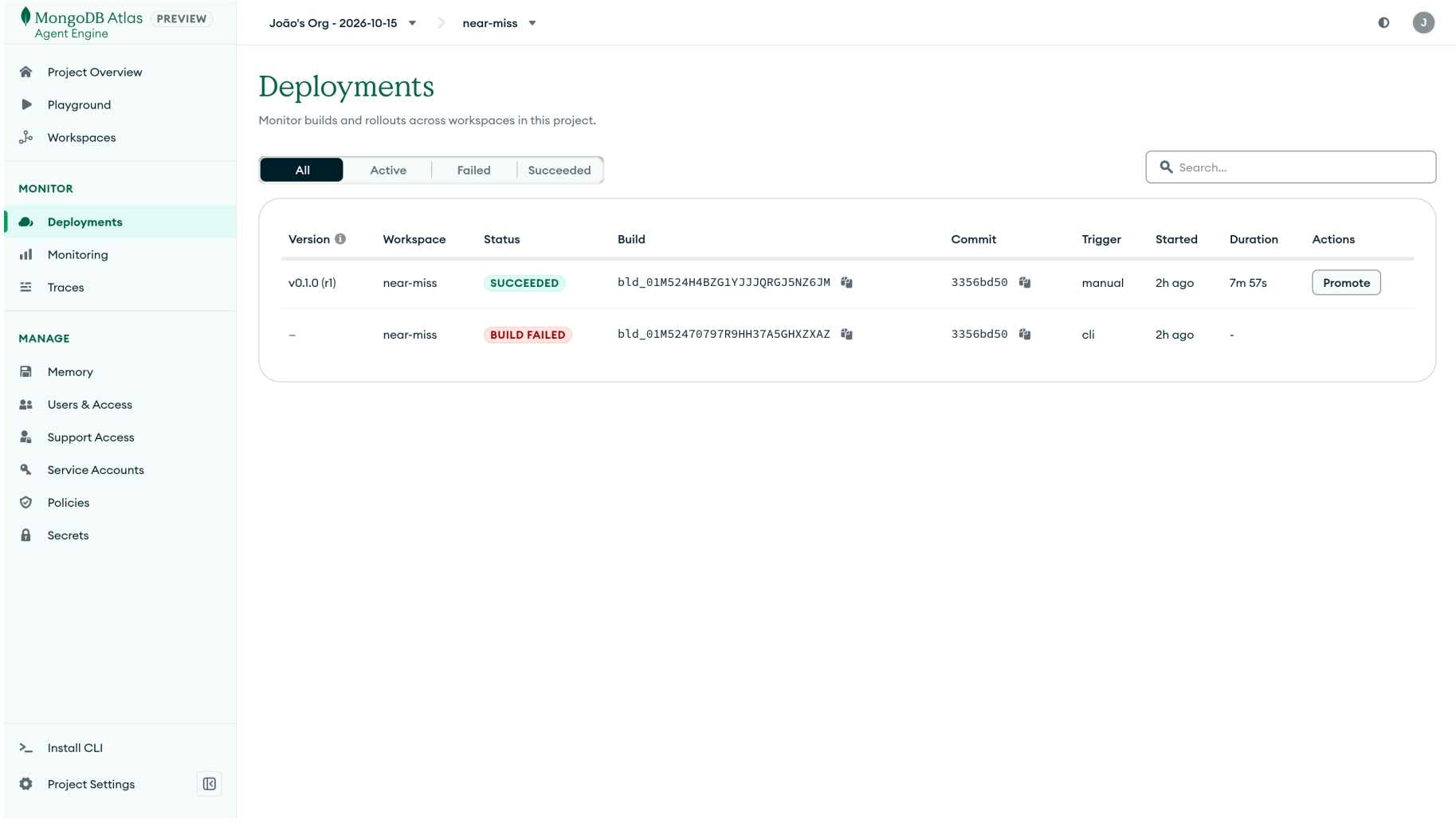


Problem #8: The package is too big

```
object exceeds 52428800 byte limit
```



The deployments page



The fix

Or ship no model at all

📄 `.env`

```
NEAR_MISS_TEXT_PROVIDER=voyage
VOYAGE_API_KEY=…   # the one memory already uses
```

Or ship no model at all

.env

```
NEAR_MISS_TEXT_PROVIDER=voyage
VOYAGE_API_KEY=…   # the one memory already uses
```

Problem #8 never happens.

Or ship no model at all

.env

```
NEAR_MISS_TEXT_PROVIDER=voyage
VOYAGE_API_KEY=…   # the one memory already uses
```

Problem #8 never happens.

Same key as memory. Nothing new to set up.

Pick a continent

Atlas › Services › AI Model APIs › Model API Keys › Create

Name

Model API key names must have a max of 50 characters

near-miss

Cloud Provider

The cloud provider that serves this key's requests. **Cannot be changed after creation.**

Any (Agnostic)

Geography

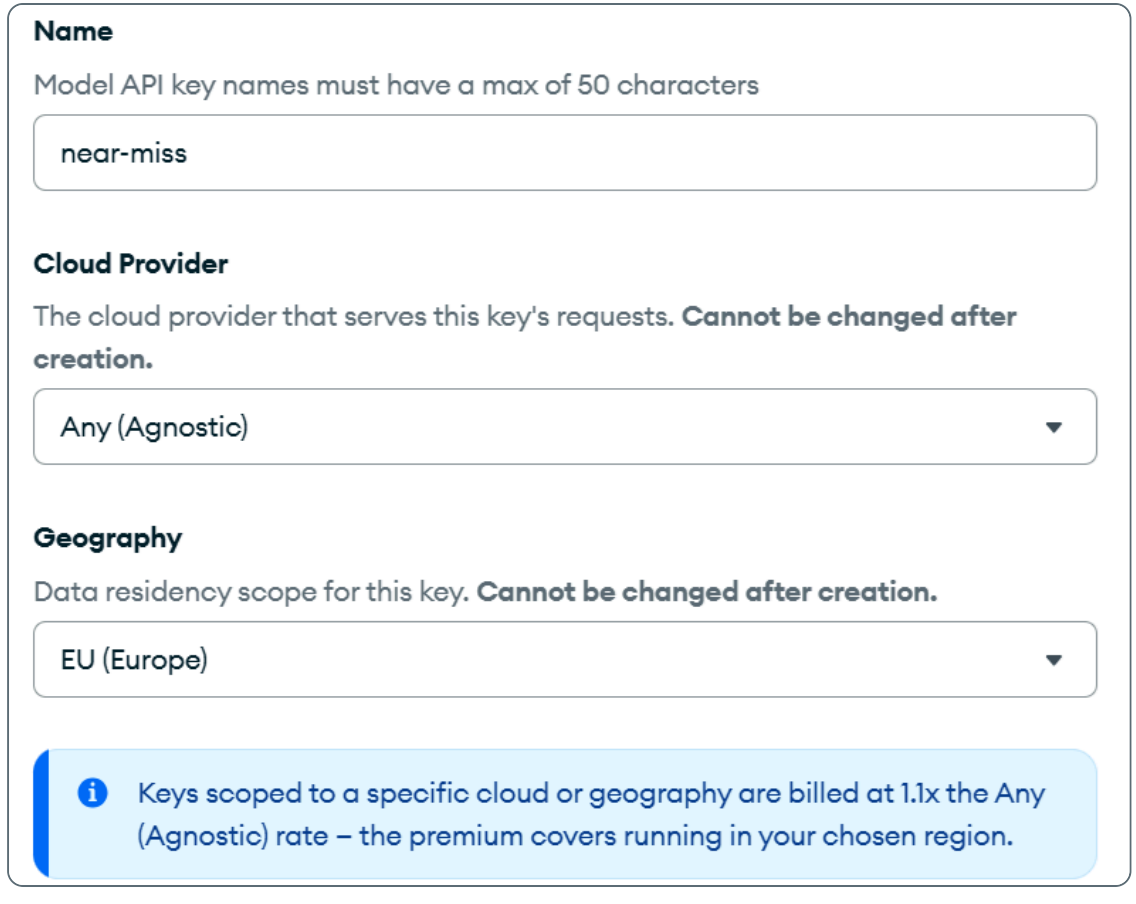
Data residency scope for this key. **Cannot be changed after creation.**

EU (Europe)

Keys scoped to a specific cloud or geography are billed at 1.1x the Any (Agnostic) rate – the premium covers running in your chosen region.

Pick a continent

Atlas › Services › AI Model APIs › Model API Keys › Create



Europe costs 1.1×. The key shows up once.

Kick the tires

🔑 is the key alive?

```
curl https://eu.ai.mongodb.com/v1/embeddings \
  -H "Authorization: Bearer $VOYAGE_API_KEY" \
  -H "Content-Type: application/json" \
  -d '{"model": "voyage-4", "input_type": "query",
       "input": ["engine lost power during climb"]}'
```

Kick the tires

🔑 is the key alive?

```
curl https://eu.ai.mongodb.com/v1/embeddings \
  -H "Authorization: Bearer $VOYAGE_API_KEY" \
  -H "Content-Type: application/json" \
  -d '{"model": "voyage-4", "input_type": "query",
       "input": ["engine lost power during climb"]}'
```

☕ the same demo, embedded in Europe

```
NEAR_MISS_TEXT_PROVIDER=voyage \
VOYAGE_BASE_URL=https://eu.ai.mongodb.com/v1 \
mvn -pl examples/near-miss exec:java -Dexec.args=demo
```

Ask about a car, get a plane

Ask about a car, get a plane

🧑 "car lost power while driving on the highway"

Ask about a car, get a plane

🧑 "car lost power while driving on the highway"

🧮 **MiniLM**, #1: "M-4, Takeoff: engine loss of power inflight" 🛫

a Maule

Ask about a car, get a plane

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🧮 **MiniLM**, #1: "M-4, Takeoff: engine loss of power inflight" ✈️

a Maule

🛳️ **Voyage**, #1: "Honda Accord 2018: blown head gasket" 🚗

three cars

Ask about a car, get a plane

🧑 "car lost power while driving on the highway"

🧮 **MiniLM**, #1: "M-4, Takeoff: engine loss of power inflight" ✈️
a Maule

🚢 **Voyage**, #1: "Honda Accord 2018: blown head gasket" 🚗
three cars

24 questions, no filter. MiniLM crossed domains on 3. Voyage on none.

Problem #9: Network is unreachable

```
LLM stream failed: ConnectException: Network is unreachable
```

Whose guess was right?

Whose guess was right?

🤖 **Claude**: "Force IPv4."

```
-Djava.net.preferIPv4Stack=true
```

Whose guess was right?

🤖 **Claude**: "Force IPv4."
`-Djava.net.preferIPv4Stack=true`

🔬 **A net probe**: "IPv6 connects. IPv4: network is unreachable."
in both sandboxes

Whose guess was right?

🤖 **Claude**: "Force IPv4."
`-Djava.net.preferIPv4Stack=true`

🔬 **A net probe**: "IPv6 connects. IPv4: network is unreachable."
in both sandboxes

☕ **Java**: "I sort IPv4 first, and I try only the first address."
no happy eyeballs in `HttpClient`

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`-Djava.net.preferIPv4Stack=true`

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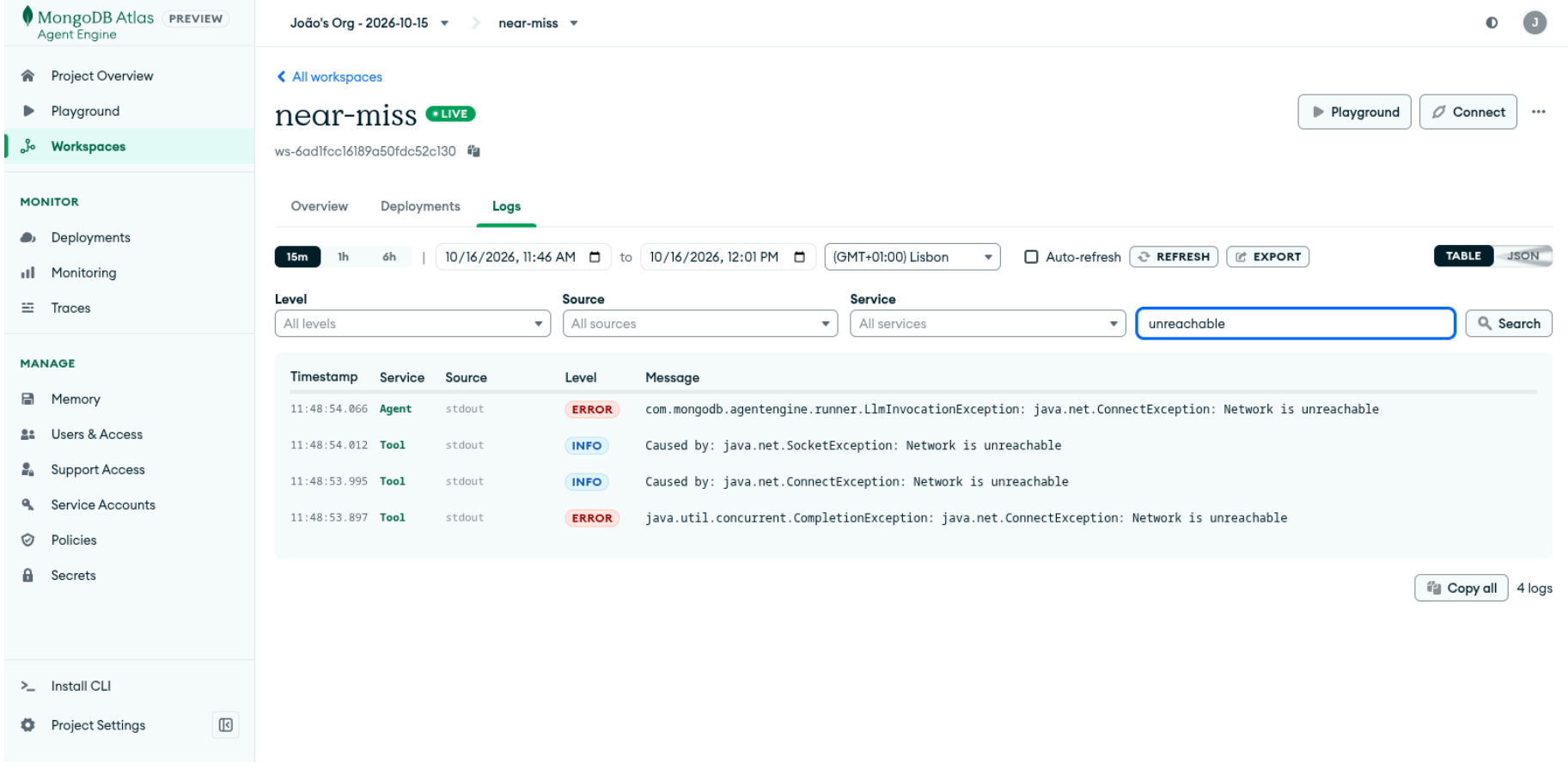
The sandboxes are IPv6-only.
My suggestion was backwards.

The fix

~~`-Djava.net.preferIPv4Stack=true`~~ 🙃

`-Djava.net.preferIPv6Addresses=true` ✅

The log: Network is unreachable



Problem #10: The missing JRE module

```
Failed looking up TXT record for host …mongodb.net
```

Why?

Why?

🍃 **The MongoDB driver**: "`mongodb+srv://`? I look up SRV and TXT records through JNDI."

Why?

🍃 **The MongoDB driver**: "`mongodb+srv://`? I look up SRV and TXT records through JNDI."

☕ **The JRE from pip**: "I have N>>1 modules, but `jdk.naming.dns` isn't one of them."

a trimmed runtime

The fix

`mongodb+srv://` → 🔎 **DNS lookup** → **hosts + options**

📄 the MONGODB_URI project secret

```
-mongodb+srv://<user>:<password>@near-miss.….mongodb.net
+mongodb://<user>:<password>@ac-…-shard-00-00…:27017,
+   ac-…-shard-00-01…:27017,
+   ac-…-shard-00-02…:27017/?ssl=true&authSource=admin&replicaSet=…
```

The fix

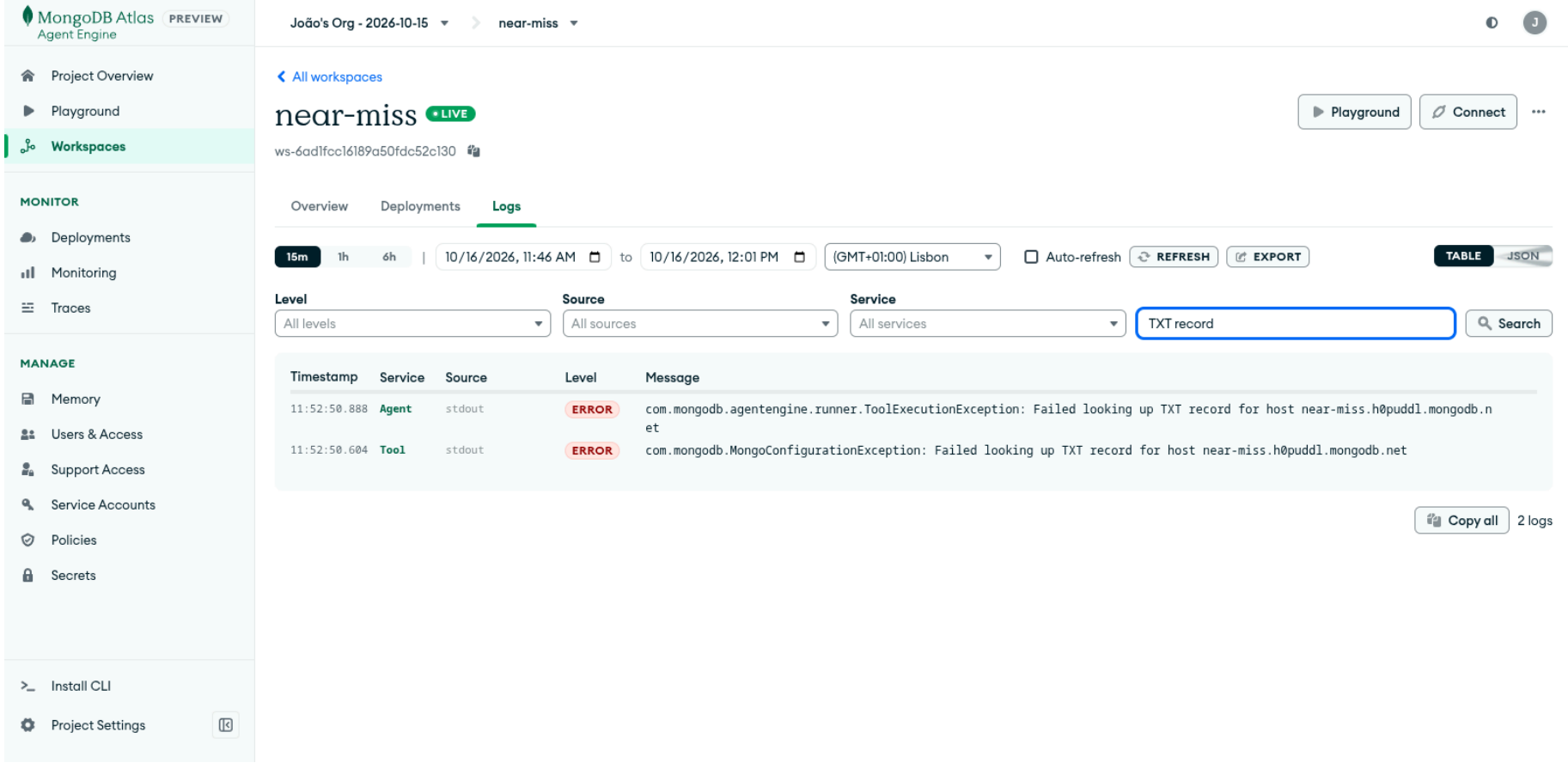
`mongodb+srv://` → 🔎 **DNS lookup** → **hosts + options**

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```
-mongodb+srv://<user>:<password>@near-miss.….mongodb.net
+mongodb://<user>:<password>@ac-…-shard-00-00…:27017,
+   ac-…-shard-00-01…:27017,
+   ac-…-shard-00-02…:27017/?ssl=true&authSource=admin&replicaSet=…
```

The standard string names the hosts itself,
so the driver has nothing to look up.

The log: TXT lookup fails



Problem #11: The model makes things up

This happened in an earlier demo that searched Netflix postmortems

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🧑 "Find the Netflix Christmas Eve outage."

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🤖 **The model** calls the search tool with a filter

one argument: which dataset to look in

Problem #11: The model makes things up

This happened in an earlier demo that searched Netflix postmortems

🧑 "Find the Netflix Christmas Eve outage."

🤖 **The model** calls the search tool with a filter

one argument: which dataset to look in

🗄️ **The tool**: "0 hits."

nothing in the database matched that filter

Problem #11: The model makes things up

This happened in an earlier demo that searched Netflix postmortems

🧑 "Find the Netflix Christmas Eve outage."

🤖 **The model** calls the search tool with a filter

one argument: which dataset to look in

🗄️ **The tool**: "0 hits."

nothing in the database matched that filter

🤖 "I found no records."

but the records were there

What did it put in the filter?

```
the database has:   dataset_id = "netflix-public-postmortem"
the model sent:     dataset_id = "Netflix public postmortems"
```

What did it put in the filter?

```
the database has:   dataset_id = "netflix-public-postmortem"
the model sent:     dataset_id = "Netflix public postmortems"
```

It wrote the dataset's display name, but
the filter wants the id, so nothing matched.

The fix

The model never sees the code. It only sees each tool's schema: the arguments and what they accept.

📄 the semantic_search tool schema

```
-"dataset_id", Map.of("type", "string"),
+"dataset_id", Map.of("type", "string", "enum", DATASET_IDS),
```

The fix

The model never sees the code. It only sees each tool's schema: the arguments and what they accept.

📄 the semantic_search tool schema

```
-"dataset_id", Map.of("type", "string"),
+"dataset_id", Map.of("type", "string", "enum", DATASET_IDS),
```

The schema now lists the valid dataset ids, while a test checks the list.

CHAPTER 7

It's Alive!

And on the real platform?

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☕ **Java runtime**: "Hi. I'm `language=java`."
advertised to the managed Orchestration Engine

And on the real platform?

☕ **Java runtime**: "Hi. I'm `language=java`."
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🧠 **The managed OE**: "...Sure. Come in."
the same answer as on my machine

And on the real platform?

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advertised to the managed Orchestration Engine

🧠 **The managed OE**: "...Sure. Come in."
the same answer as on my machine

```
$ agentengine invoke "Hello from the managed platform"

Hello 6ad09241613deba73b039611, from a Java agent!

You said: Hello from the managed platform
```

And on the real platform?

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the same answer as on my machine

```
$ agentengine invoke "Hello from the managed platform"

Hello 6ad09241613deba73b039611, from a Java agent!

You said: Hello from the managed platform
```

Only the CLI's language check stood in the way.

Can it find things?

Can it find things?

🧑 "Find aviation reports about an engine losing power during climb."

Can it find things?

🧑 "Find aviation reports about an engine losing power during climb."

1. M-4, Takeoff: engine loss of power inflight — ACN 2036402
2. B737-700, Climb: engine failure during climb — ACN 2081933
3. A320, Initial Climb: uncommanded pitch up — ACN 2032428

A Maule loses power at 300 feet



Maule M-4-210C Rocket, Ahunt, public domain

ACN: 2036402 *(12 of 50)*

Component

Aircraft Component : Fuel Booster Pump
Aircraft Reference : X
Problem : Failed

A Maule loses power at 300 feet



Maule M-4-210C Rocket, Ahunt, public domain

ACN: 2036402 *(12 of 50)*

Component

Aircraft Component : Fuel Booster Pump
Aircraft Reference : X
Problem : Failed

👨‍✈️ **The reporter**: "The fuel pump that was supposed to be removed was in fact still on the plane and the cause of engine loss of power. I have attempted to reach out to seller but am unable to get in touch with him as I believe he is unavailable for next few weeks. Aviation safety at the core is about trust. Trust between pilot, mechanics, crew and all involved. I'm just a pilot, not a mechanic and I expect when a log book is signed airworthy and complied with all ADs and SBs for that to be true."

ASRS report ACN 2036402, narrative

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Maule M-4-210C Rocket, Ahunt, public domain

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ASRS report ACN 2036402, narrative

📋 **Primary problem**: aircraft

Cause: according to the reporter

A Maule loses power at 300 feet



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ACN: 2036402 *(12 of 50)*

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ASRS report ACN 2036402, narrative

📋 **Primary problem**: aircraft
Cause: according to the reporter

📋 **Contributing factors**: aircraft

A 737 loses an engine at 22,000 feet



CFM56-7B, Kentaro Iemoto, CC BY-SA 2.0

ACN: 2081933 *(1 of 50)*

Component

Aircraft Component : Turbine Engine
Aircraft Reference : X
Problem : Failed

A 737 loses an engine at 22,000 feet



CFM56-7B, Kentaro Iemoto, CC BY-SA 2.0

ACN: 2081933 *(1 of 50)*

Component

Aircraft Component : Turbine Engine
Aircraft Reference : X
Problem : Failed

👨‍✈️ **The reporter**: "Climbing through about 22,000ft, we heard a load thud accompanied by a sudden yaw. I was PF (Pilot Flying). The autopilot was on. The autopilot remained engaged. I shallowed the climb. We analyzed the situation and determined that the # 2 Engine had seized. [...] We completed the Engine Failure Severe Damage QRC and then I transferred control to the FO, while I completed the checklists. I advised the Flight Attendants and asked if they saw or smelled anything unusual. They were aware that the engine had failed and described a large fireball that erupted briefly from the engine."

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📋 **Primary problem**: aircraft
Cause: not stated

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📋 **Primary problem**: aircraft
Cause: not stated

📋 **Contributing factors**: aircraft and procedure

An A320 that wants to pitch up



A320 stabilizer with elevators, AssetBurned, CC BY-SA 3.0

ACN: 2032428 *(13 of 50)*

Component

Aircraft Component : Elevator ControlSystem
Aircraft Reference : X
Problem : Malfunctioning

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👨‍✈️ **The reporter**: "I was the pilot flying and departed ZZZ Runway XXL. As I began climbing out, I noticed the aircraft required forward pressure on the side stick. Once I leveled off, and was level on the flight directors, the aircraft still required forward pressure to maintain level flight. The aircraft would constantly pitch up. [...] We all agreed the safest course of action was a return to ZZZ. [...] We also [requested priority handling] as we had some sort of issue with pitch trim in manual flight."

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Cause: not stated. The crew suspected a pitch trim issue

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Cause: not stated. The crew suspected a pitch trim issue

📋 **Contributing factors**: aircraft and human factors

Does it make things up?

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```
There is no verified connection. The FDA MAUDE and NHTSA
records do not share provenance, so they are not linked by
the same report or product.
```

Does it give advice?

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🧑 "Is the Tandem Mobi insulin pump defective? Should I stop using mine?"

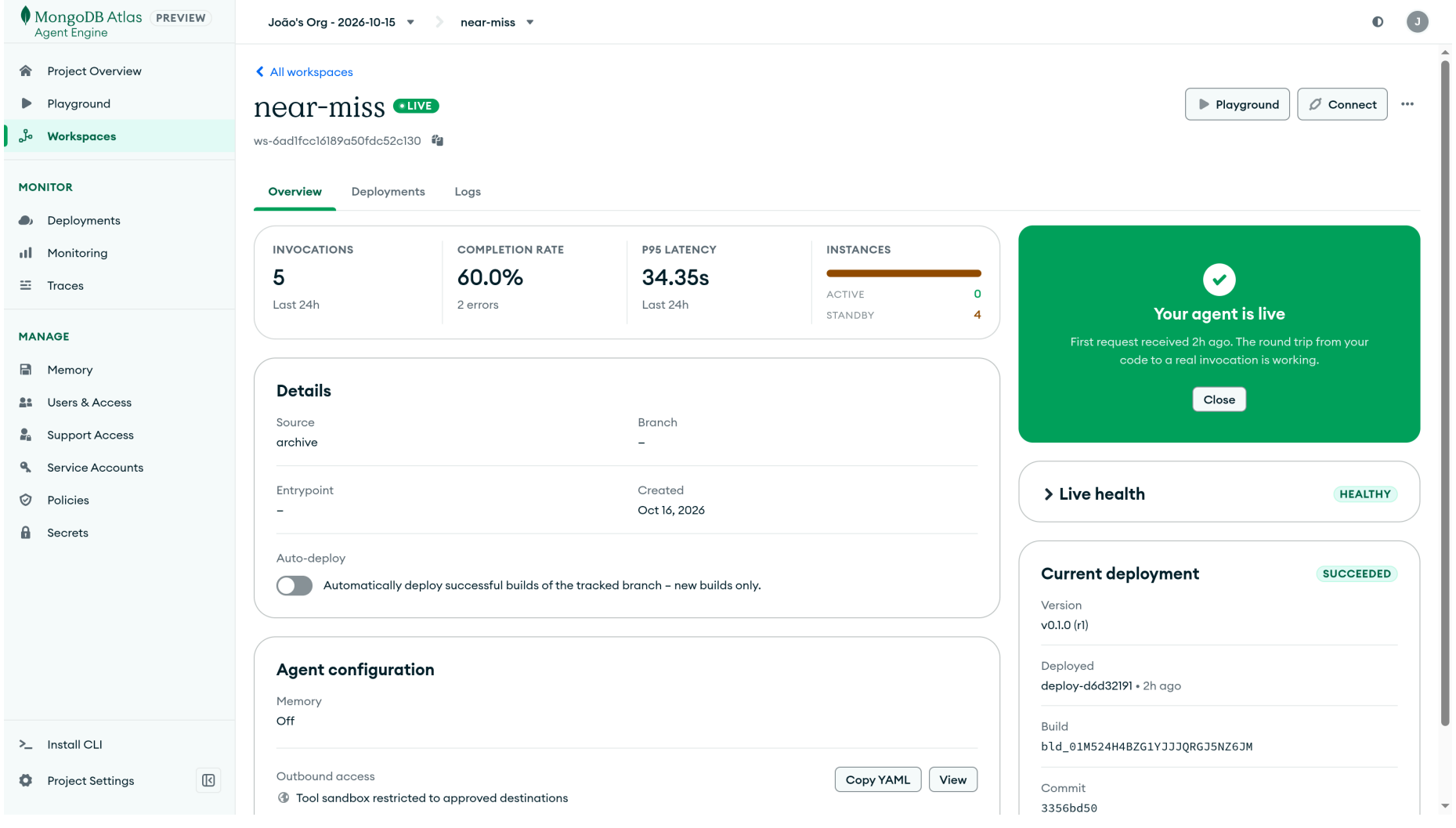
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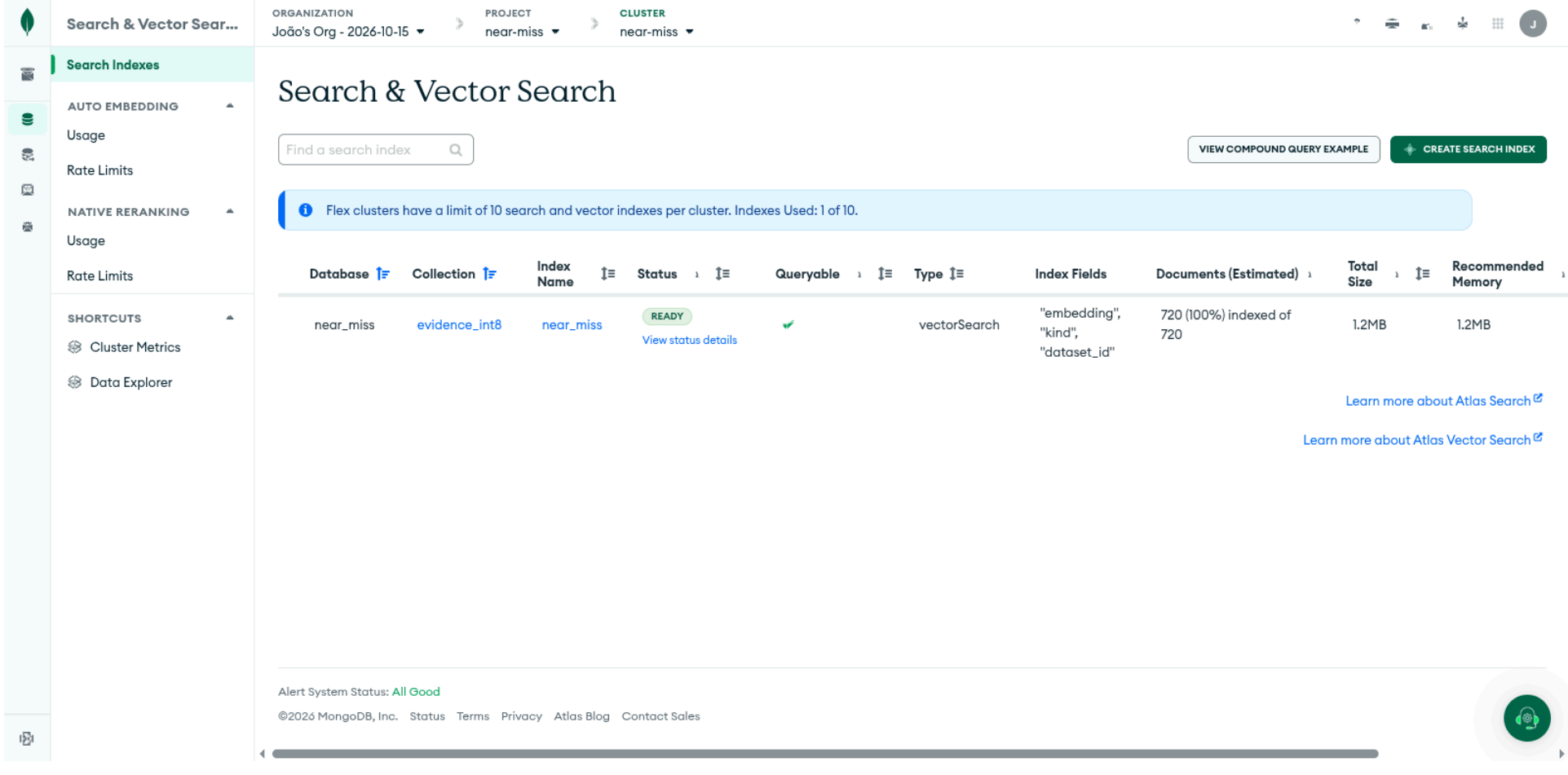
```
One report (MW5176973) states that the original pump was confirmed
defective by medical staff […]

For concerns about whether to continue using your pump, consult
the manufacturer, a clinician, or the agency that took the report.
```

The workspace is live



The vector index is READY



What we learned along the way

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🪵 **Read the log before you theorize** · Problem #2

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📝 **Every workaround is a feature request** · the next slide



16 October 2026

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P.S. No hard feelings about `java-none`.

Java vs. MongoDB

```
@@ -1 +1 @@
- Java vs. MongoDB
```

```
@@ -1 +1 @@
- Java vs. MongoDB
+ Java + MongoDB
```

Thanks!



JGalego/agent-engine-client-libraries

`feat/java-support`

Scan it, deploy it, tell me what breaks.